Today’s Agenda

- Midterm Updates
- Semester Project
- Midterm Grade Meetings
- AMNH Visit
- ML1 learning objectives
  - Glossary
  - Central Dogma table
- AlphaFold activity!
- Probability practice

Midterm Notes so far

- If the error bars are not identified explicitly as SE or 95% CI, don’t assume they are
- p-value is not the same as 95% CI
- When asked to compare results of different groups, don’t just explain the groups that are statistically significant. Include the not significant comparisons too.
  - Group A vs. Group B
  - Group A vs. Group C
  - Group B vs. Group C
- “*The p-value is less than .05 so there is a statistically significant difference*” Is NOT a sufficient comparison

Semester Project

- Sources will be approved by Saturday
- Slide Draft Due Date Nov 19th
  - Who is presenting which parts
  - General flow of the presentation
  - I will plan a meeting with each group to discuss the slides and give notes on improvements that should be made before
- Presentations will happen during our last meeting Dec 3rd
- Every group member must contribute to the slides and speak during the presentation.
- Each group will present for 20 minutes.
  - 15 minutes for presentation
  - 5 minutes for questions (1 question from other 3 groups, 2 - 4 questions from instructor = 1 question per group member))
- We will discuss group presentation strategies in November as the presentation date gets closer.

Midterm Grade Meetings

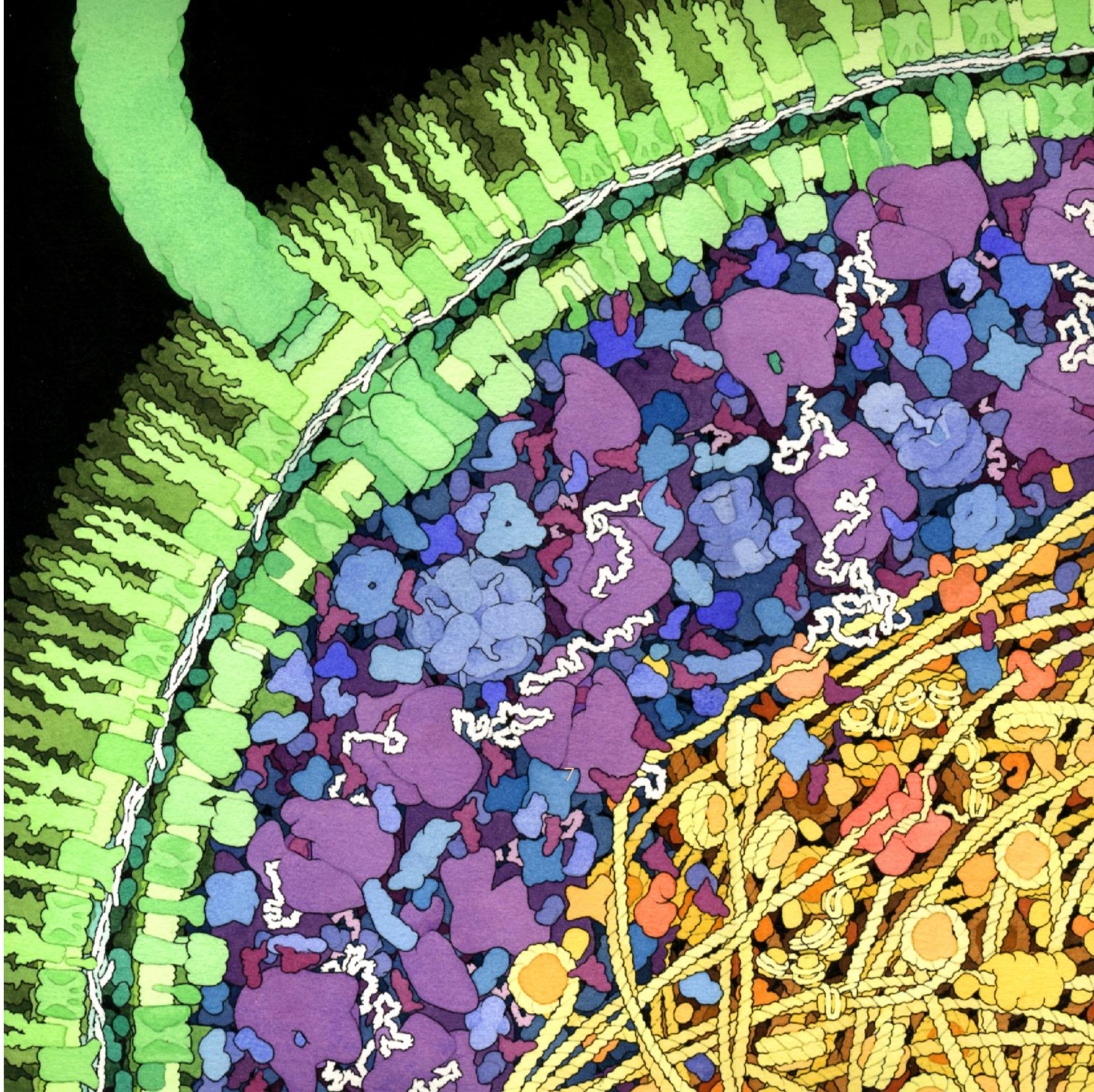
- I would like to meet with each of you individually to discuss your grades on the midterms
- Meetings will be 10-15 mins
- Week of Nov 10th
  - Most likely Thursday Nov 13th
    - Morning to evening with 15-minute time slots.

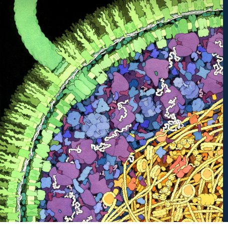
AMNH Visit, Scavenger Hunt, Selfie, & Reflection

- No lecture or seminar next week
- You are required to visit the AMNH sometime before the end of the semester
- Directions will be posted over the weekend along with the handout
- Should take around 1- 2 hrs.
- The museum will ask for a donation
  - If you would like to support the museum, you can donate what you want
  - Otherwise, you must pay 1cent minimum to enter.
- Write a reflection of your visit.
  - Which exhibits were your favorite? Why ?
  - Which exhibits were related to frontiers? How?
- I suggest you go in groups next week.
- You won’t have access to the special exhibits
- Grade will be both class participation (Scavenger Hunt) & 1 spotlight credits (Reflection)

Molecules and Life
Week 1

Chemistry of Life & Molecular Machines





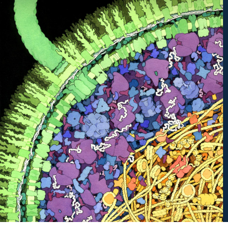
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL and AlphaFold, to access publicly available databases in efforts to understand diseases and how to combat them.



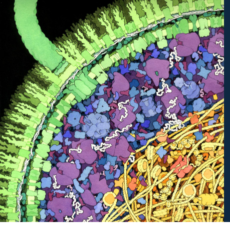
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

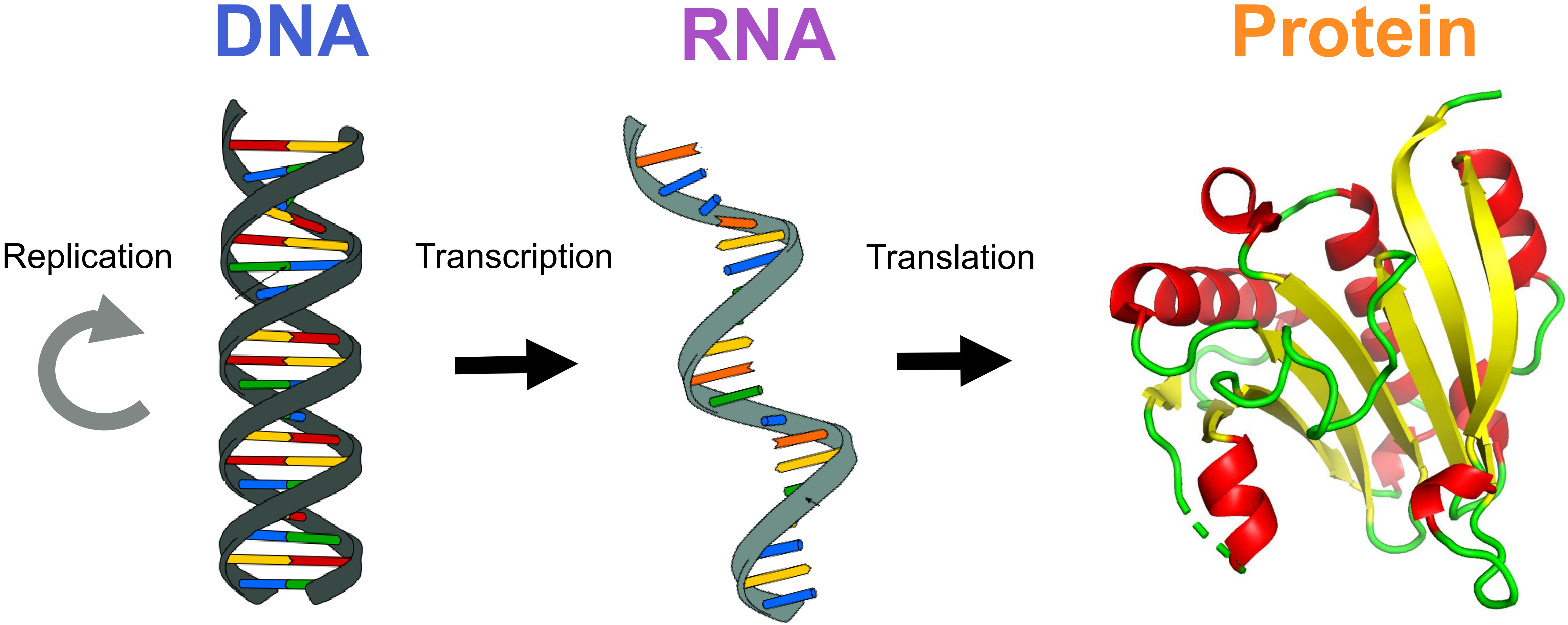
2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.

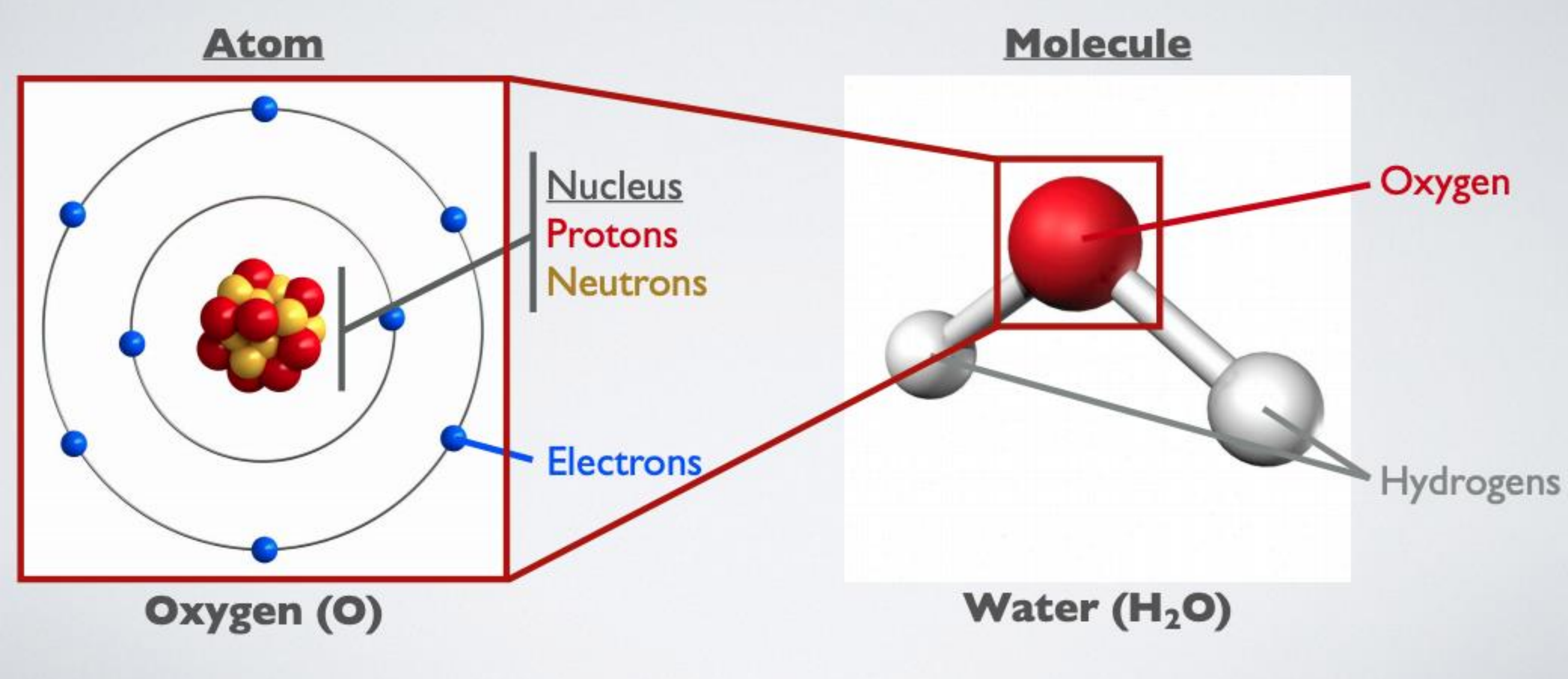


The “Central Dogma” of molecular biology



What kinds of chemical interactions do we have to think about?

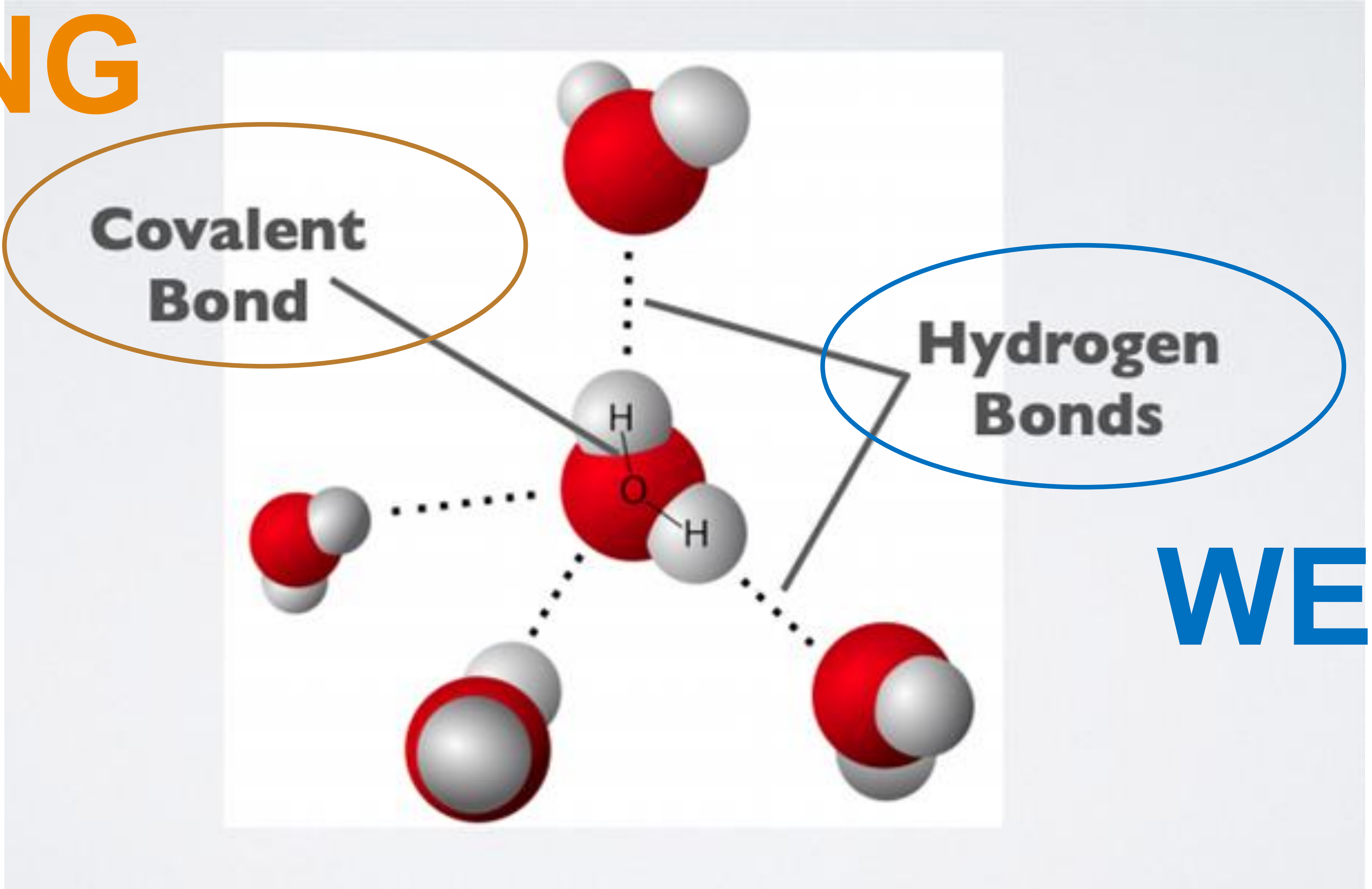
Atoms and molecules



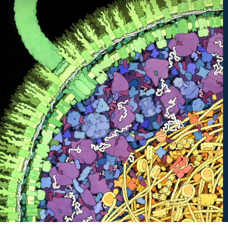
Molecule: two or more atoms bonded together

Types of bonds

STRONG



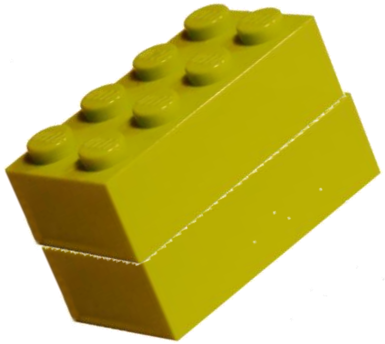
WEAK



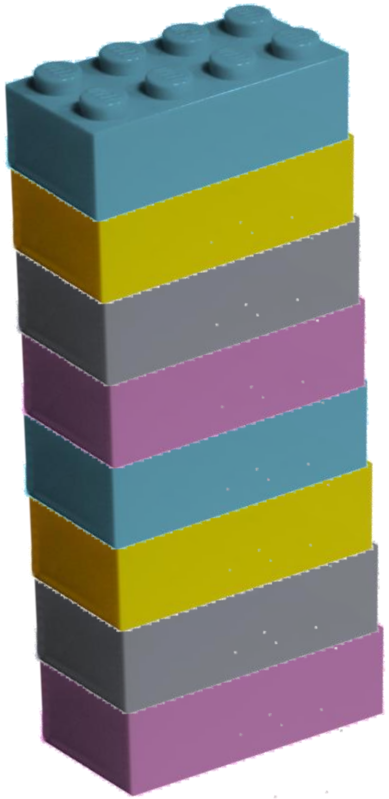
Polymers

<u>DNA, RNA, and proteins are **biopolymers**</u>

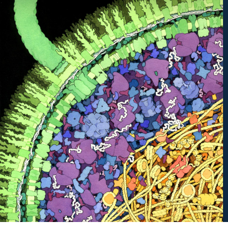
Monomers/Nucleotide /Amino Acid
small molecules which can form chemical bonds to at least two other monomer molecules.



Dimers/Dinucleotide /Dipeptide
Two identical molecules linked together



Polymers
Long repeating chains of subunits. Cannot be easily broken up due to strong covalent bonds

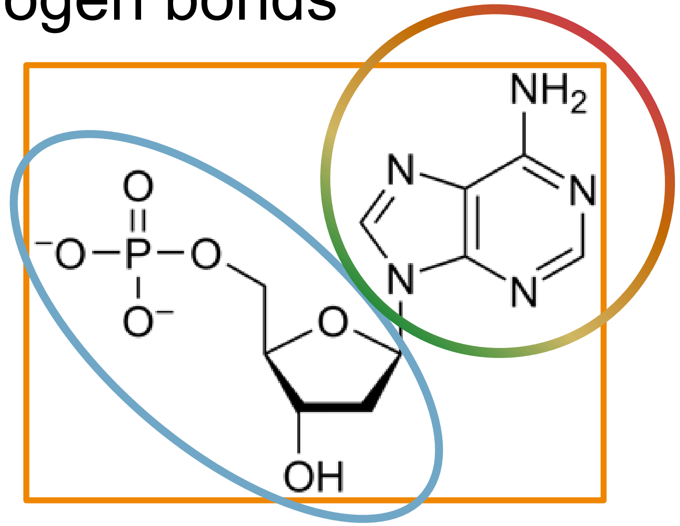


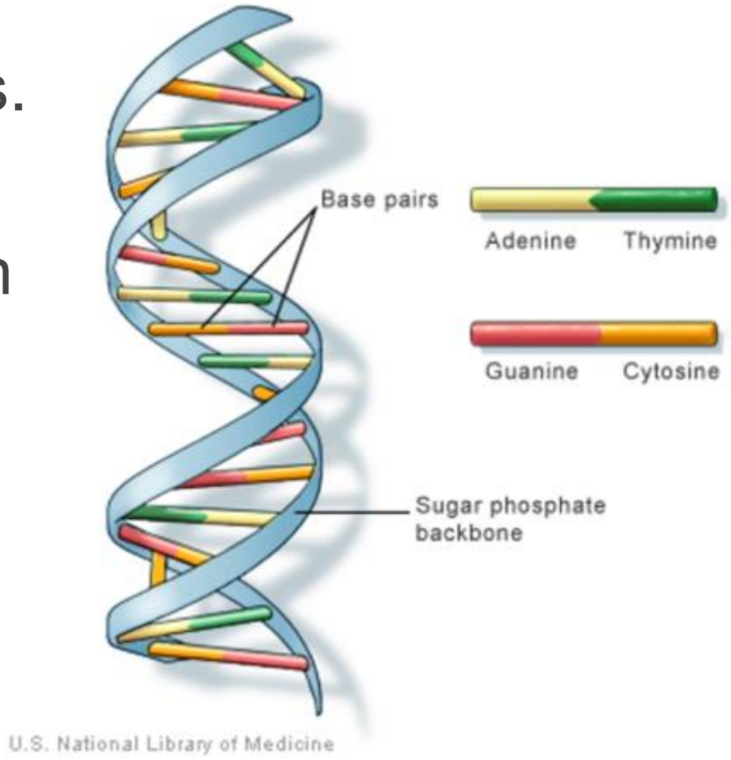
DNA: deoxyribonucleic acid

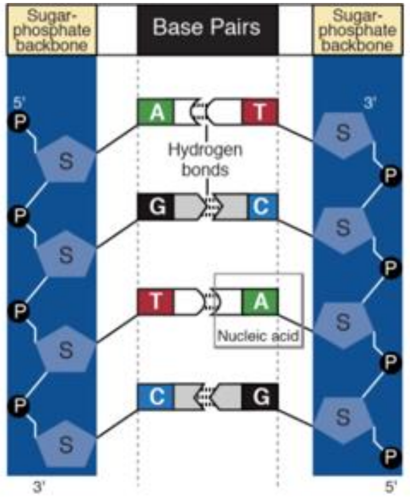
- Monomers = nucleotides
  - A, T, C, G
- Hereditary material in humans and almost all other organisms.
- Two strands wrap around each other to form a double helix
- Structure is stable and protective!

Base pairing from hydrogen bonds

Backbone from covalent bonds







RNA: ribonucleic acid

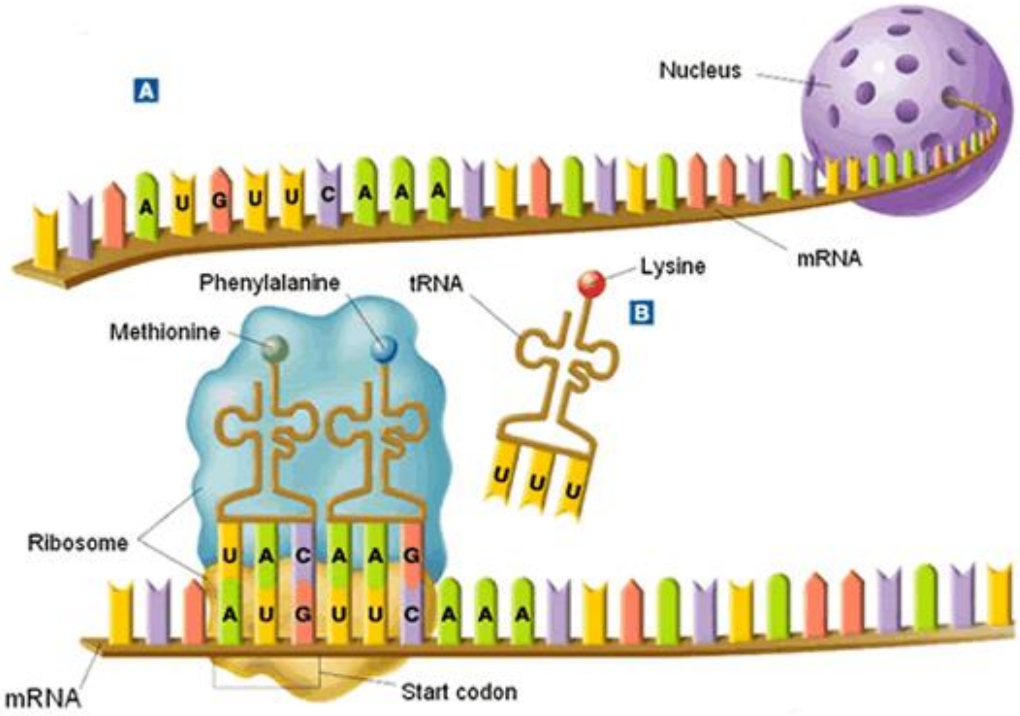
- Monomers = nucleotides
  - One different from DNA:
    - Uracil instead of Thymine

- Single-stranded

- Different types of RNA:

  - messenger RNA, or mRNA: carries information from the nucleus to ribosomes in translation.

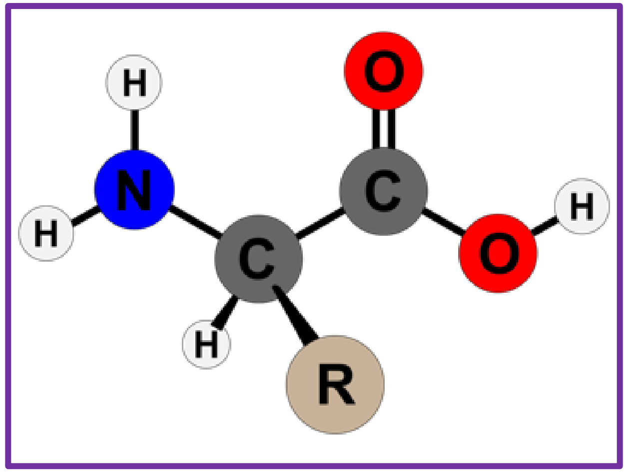
  - tRNA, or transfer RNA: physically carries amino acids to the translation site that allows them to be assembled into chains



Proteins

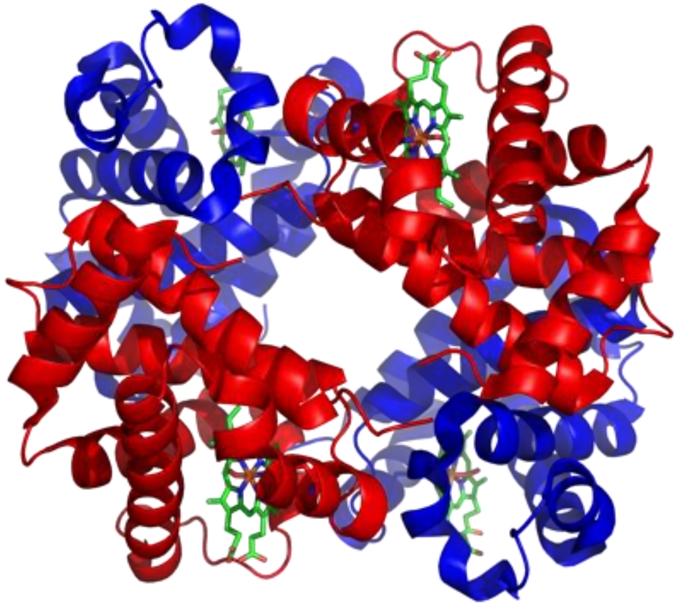
- Monomers = amino acids (20 different ones!)

- Polypeptides fold to form proteins
  - More chemical complexity with 20 amino acids
  - Covalent bonds = protein backbone
  - Hydrogen bonds = secondary (and tertiary) structure



Proteins are little machines that do jobs in the cells:

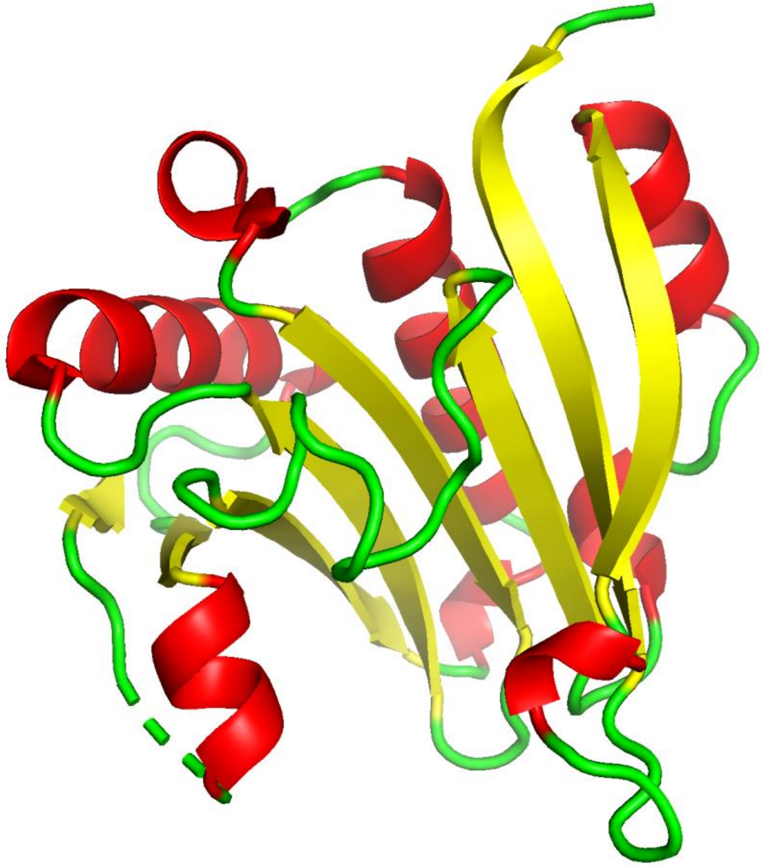
- Enzymes catalyze chemical reactions
- Some are involved in cell signaling - e.g. insulin
- Antibodies are the key players of immune system
- Receptors sense and communicate information

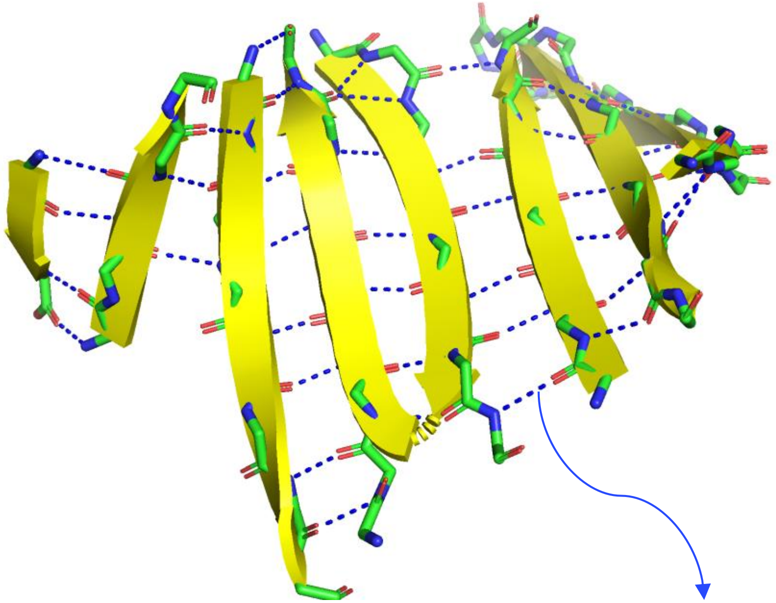


Protein secondary structures

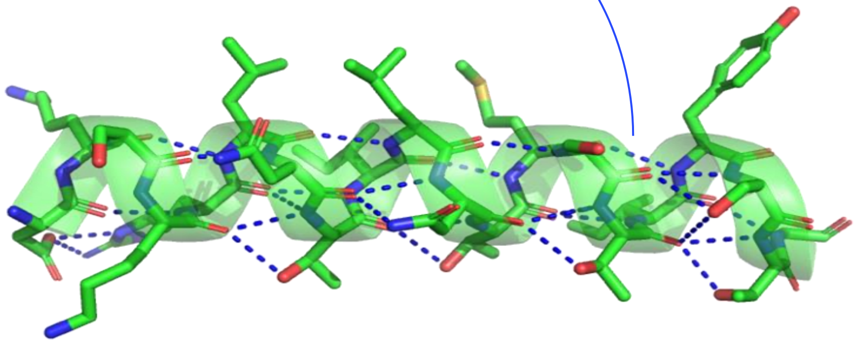
Alpha helix

Beta sheet

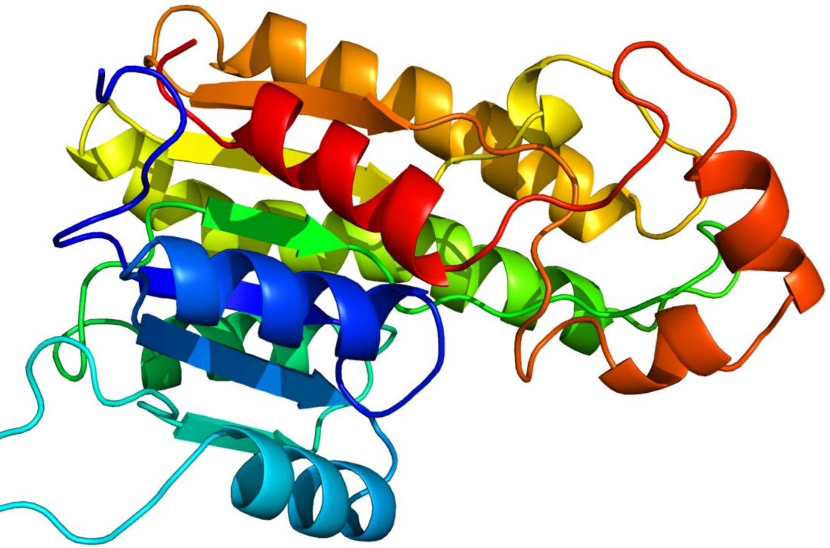
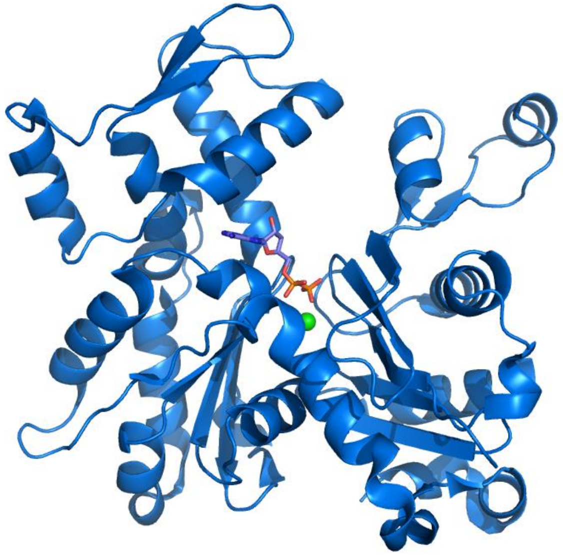
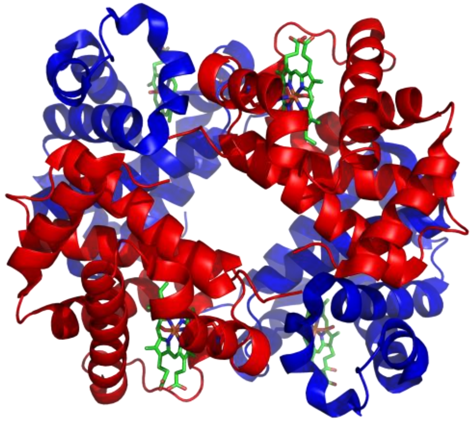
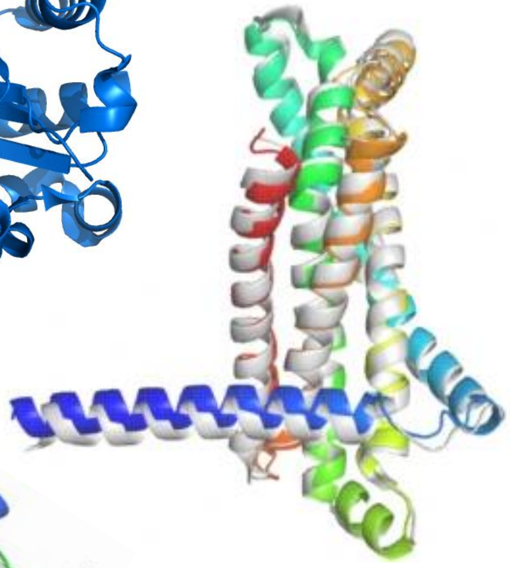
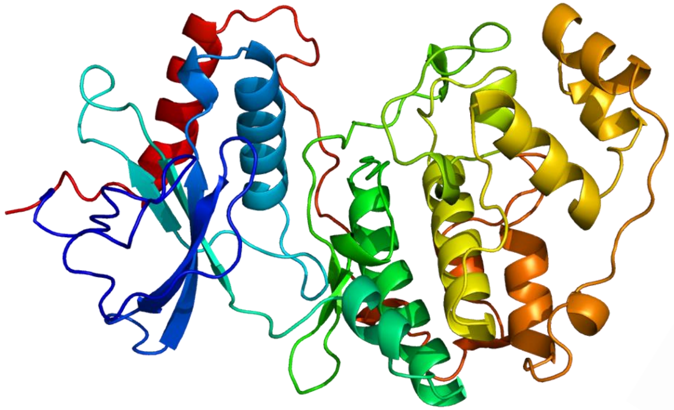
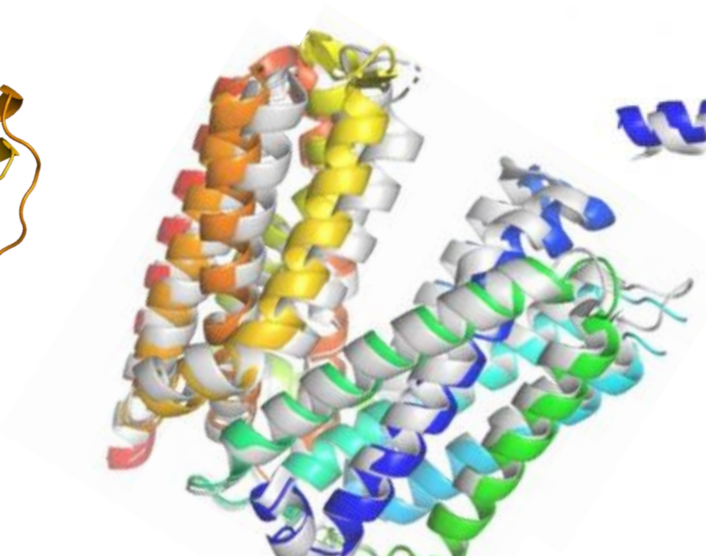
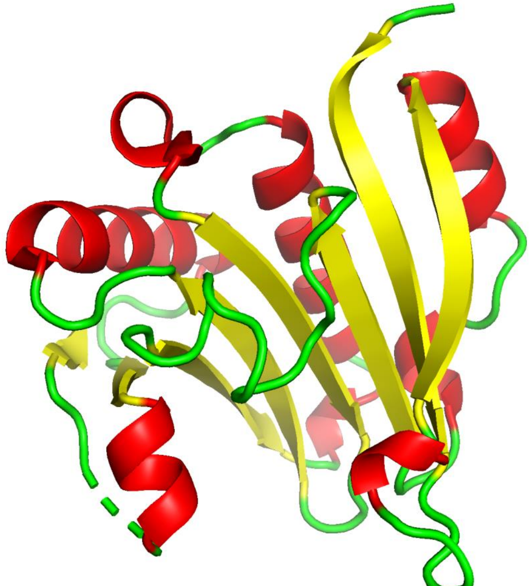


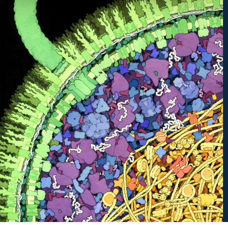


hydrogen bonds



Proteins: tiny machines



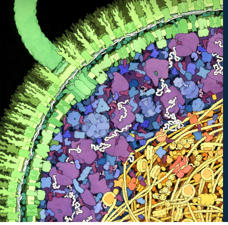
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

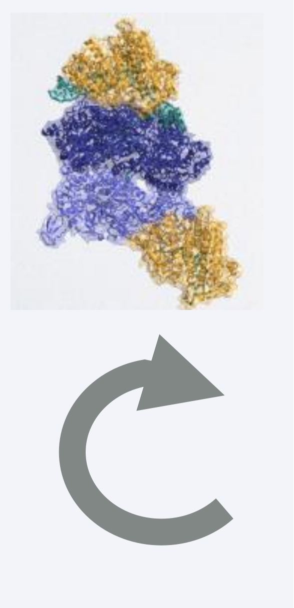
3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.

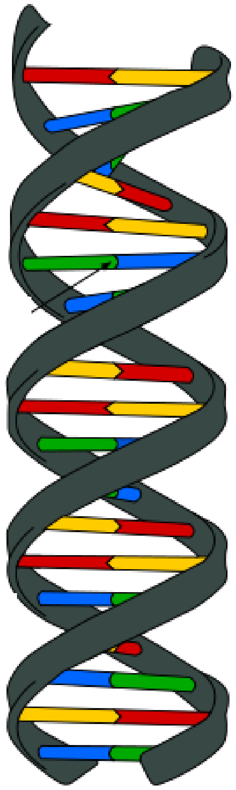


The “Central Dogma” of molecular biology

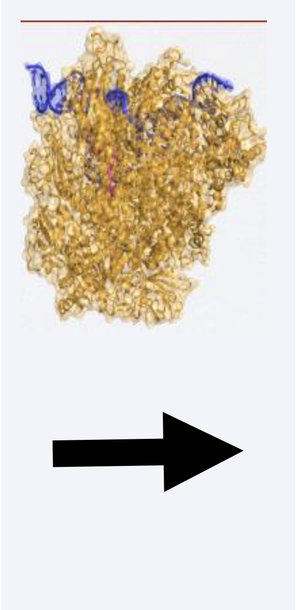
DNA Polymerase



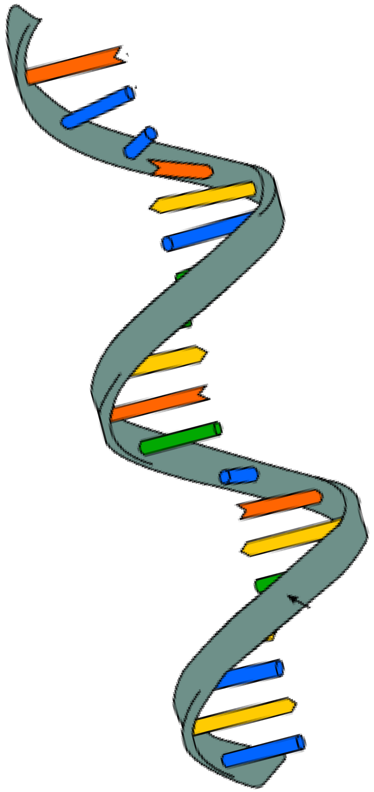
DNA



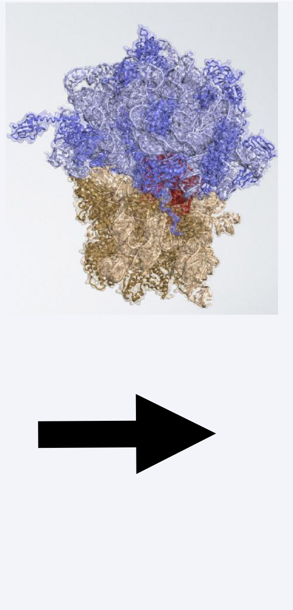
RNA Polymerase



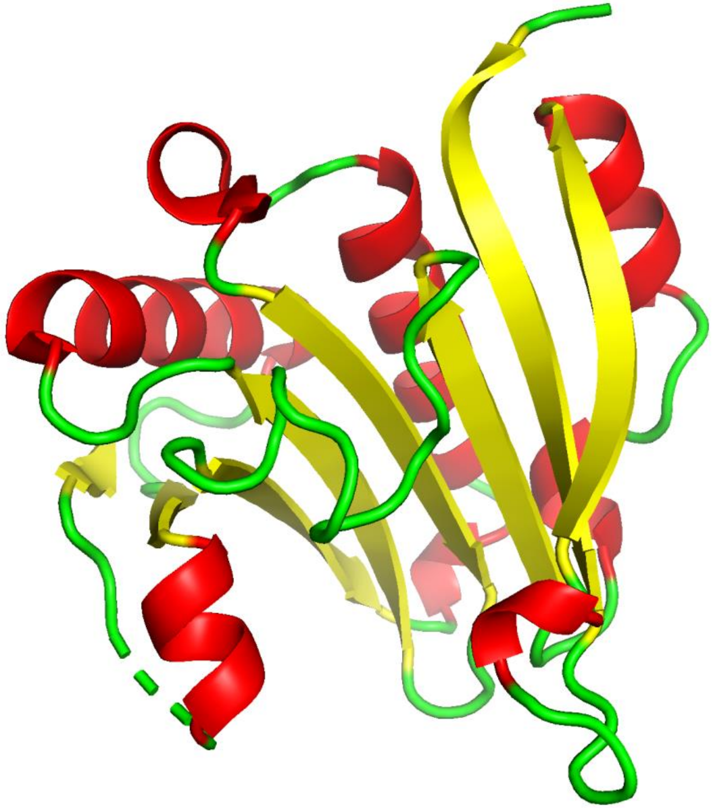
RNA



Ribosome



Protein



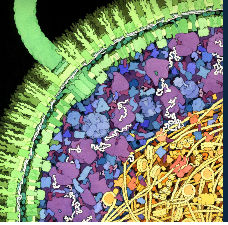
The storehouse of genetic information

Genetic messengers

The little machine

Follow along!

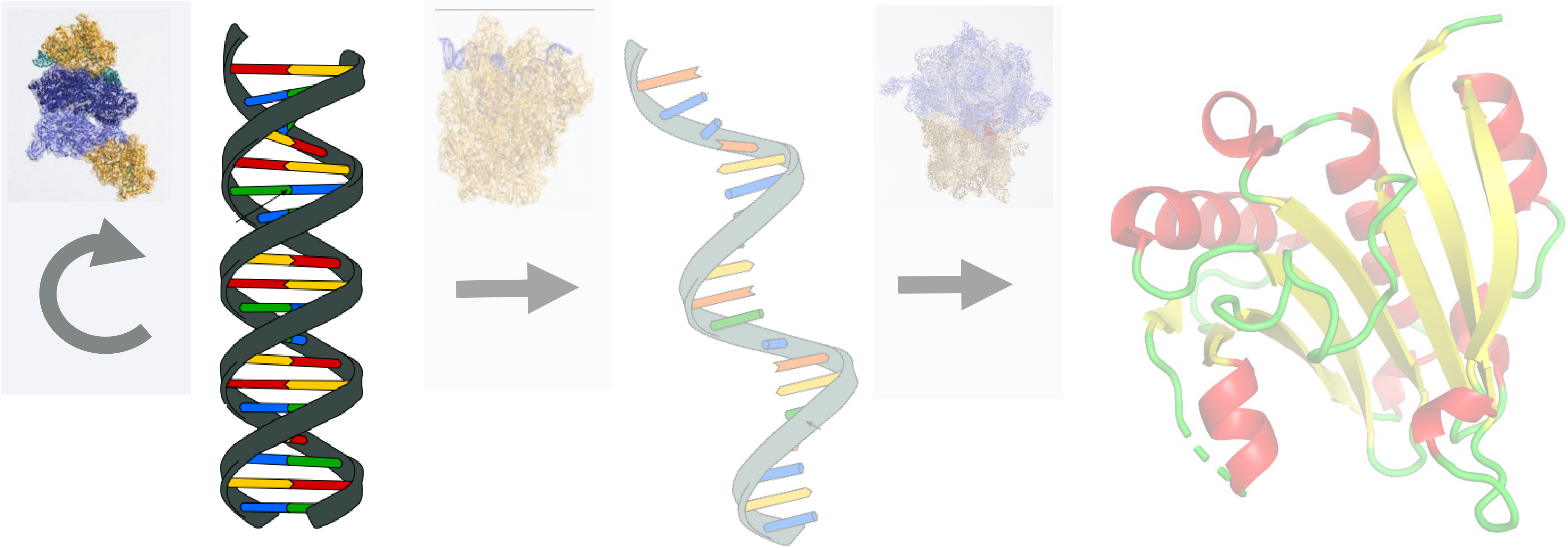
Process	Replication	Transcription	Translation
Molecular Machine			
Molecular machines are composed of…			
Input Biopolymer (= template)			
Output Biopolymer (= result of process)			
Conversion rule			
Monomers of Output			



The "Central Dogma" of molecular biology

DNA
RNA
Protein

DNA Polymerase

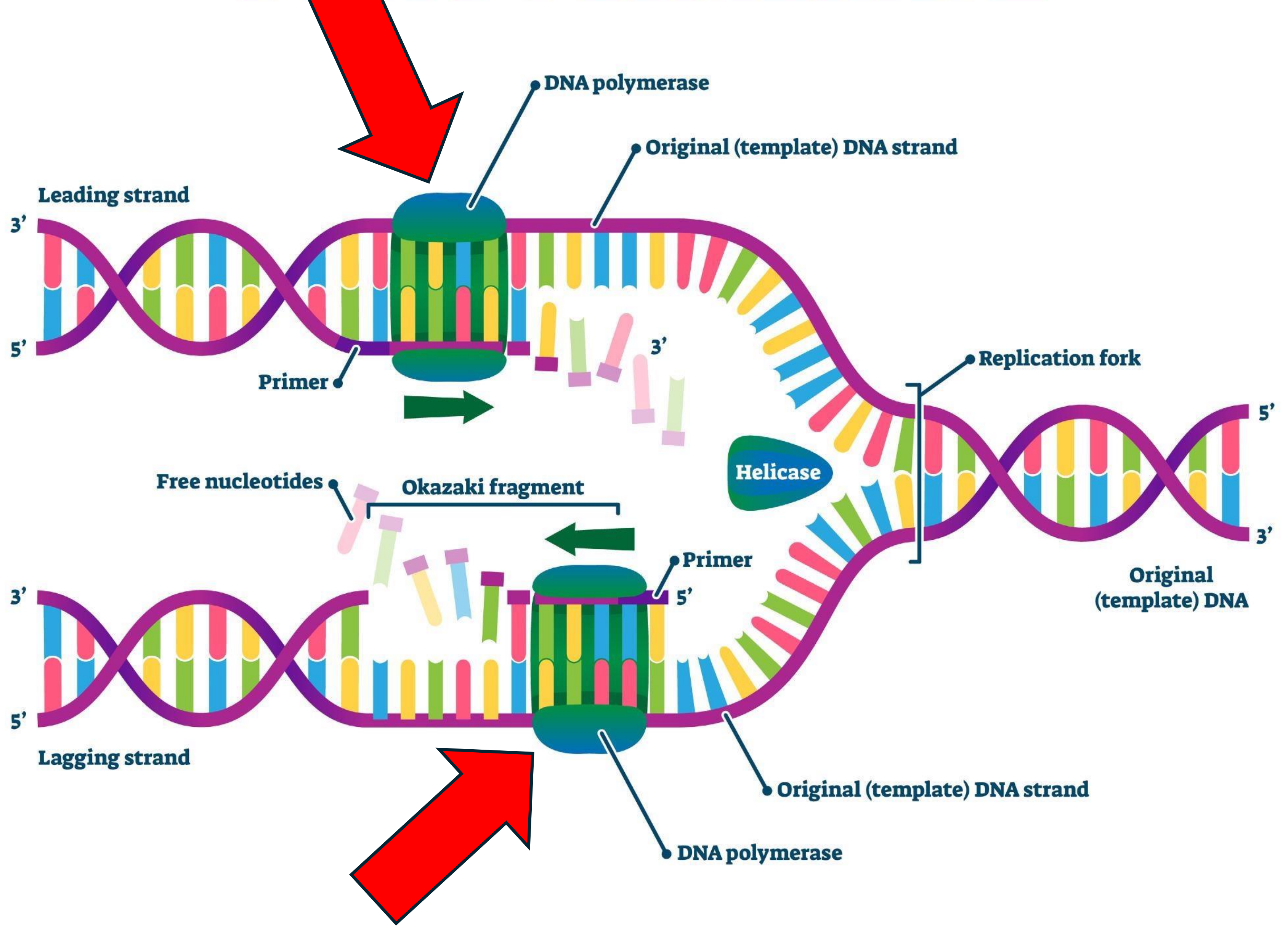


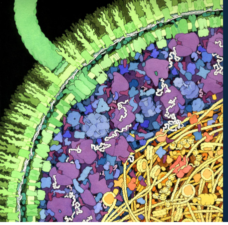
Storage of genetic info

Genetic messengers

The little machine

DNA POLYMERASE



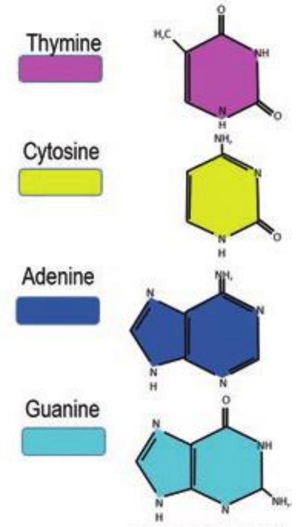


DNA replication

INPUT POLYMER

Double-stranded DNA

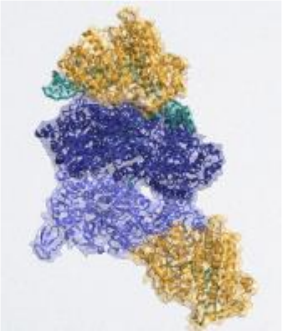
MADE UP OF



Nucleotide bases
A, C, G, T

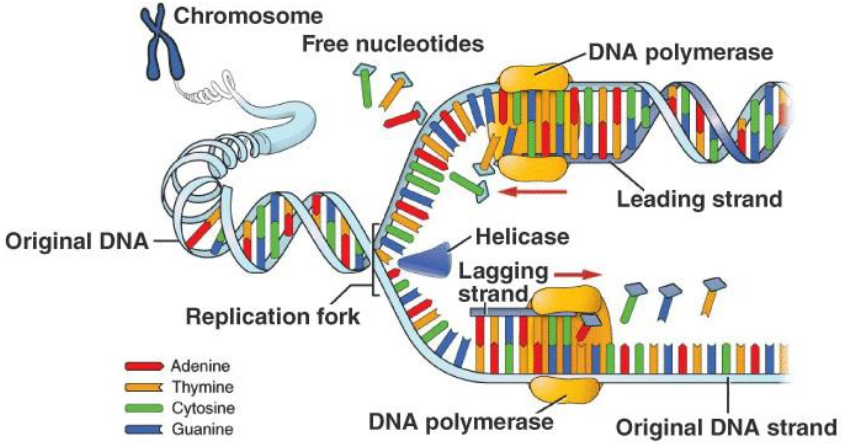
MOLECULAR MACHINE

DNA Polymerase



MADE OF

Protein
(amino acids)

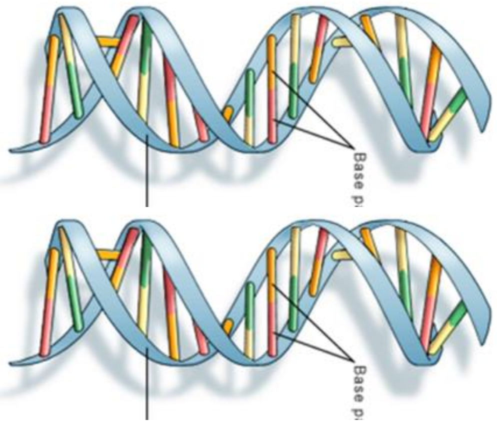


CONVERSION RULE

Base pairing (A-T and C-G)

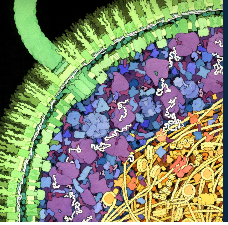
OUTPUT POLYMER

2x Double-stranded DNA



MADE UP OF

Nucleotide bases
A, C, G, T

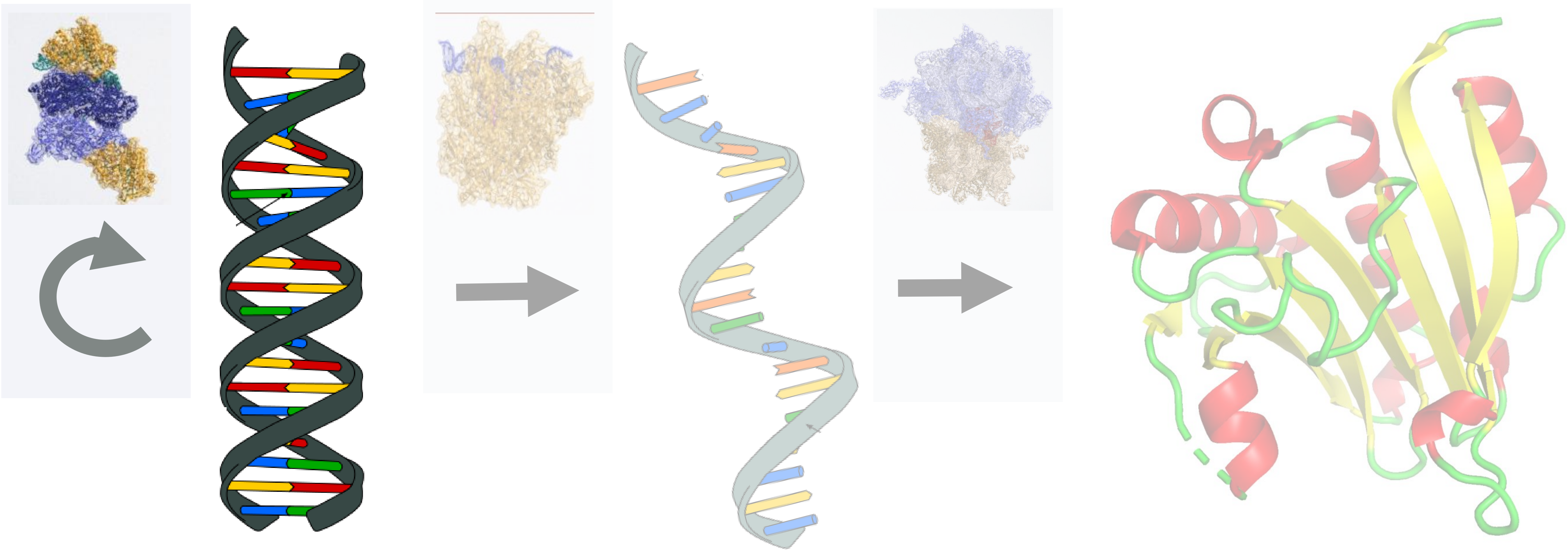


The “Central Dogma” of molecular biology

DNA

RNA

Protein



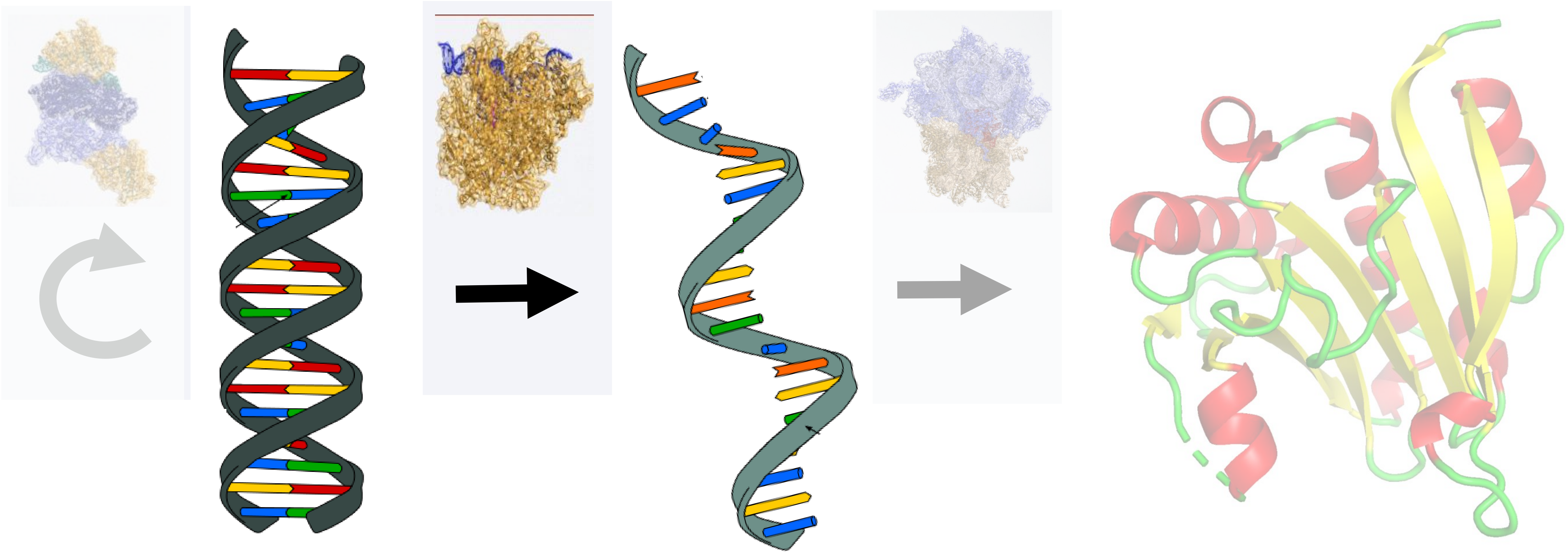
The storehouse of genetic information

Genetic messengers

The little machine

The "Central Dogma" of molecular biology

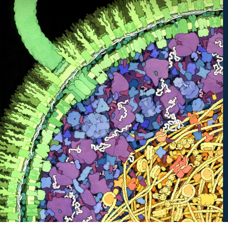
DNA **RNA** **Protein**



The storehouse of genetic information

Genetic messengers

The little machine

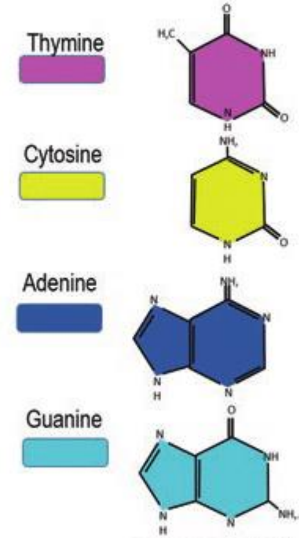


Transcription

INPUT POLYMER

Double-stranded DNA

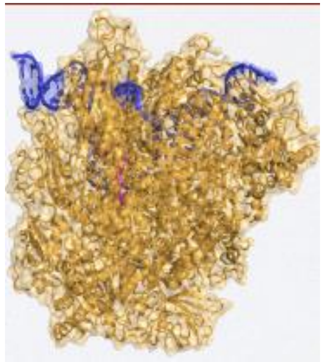
MADE UP OF



DNA
Nucleotides
A, C, G, T

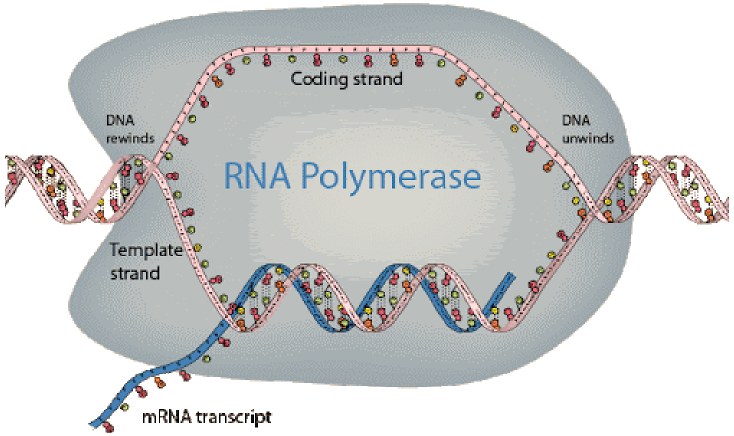
MOLECULAR MACHINE

RNA Polymerase



MADE OF

Protein
(amino acids)

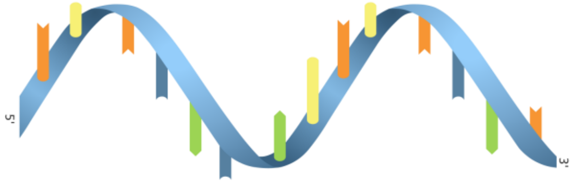


CONVERSION RULE

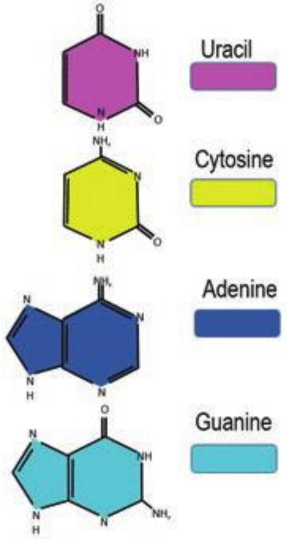
Base pairing (A-**U,** C-G, G-C, U-A)

OUTPUT POLYMER

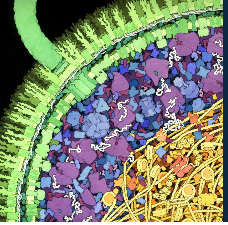
Single-stranded mRNA



MADE UP OF

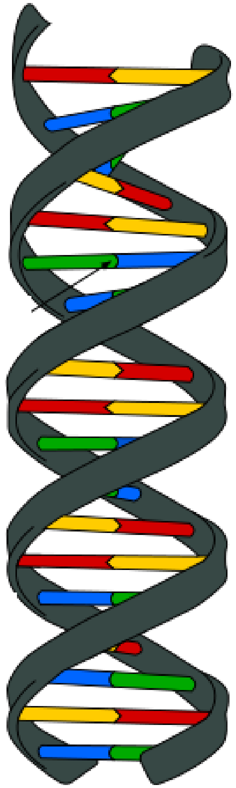


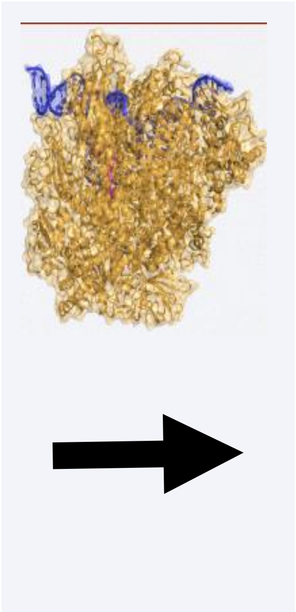
RNA nucleotides
A, C, G, U



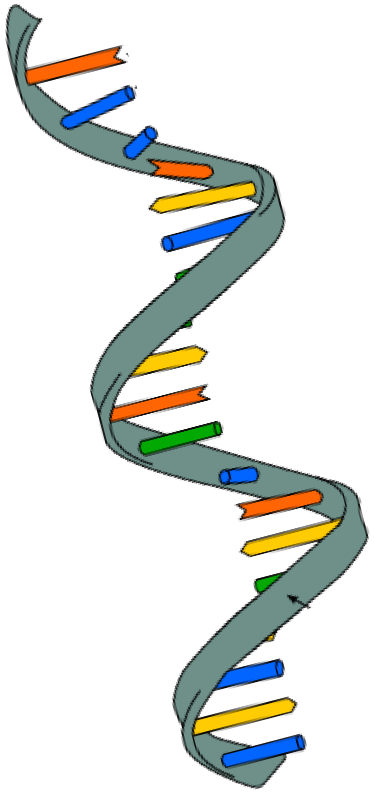
The “Central Dogma” of molecular biology

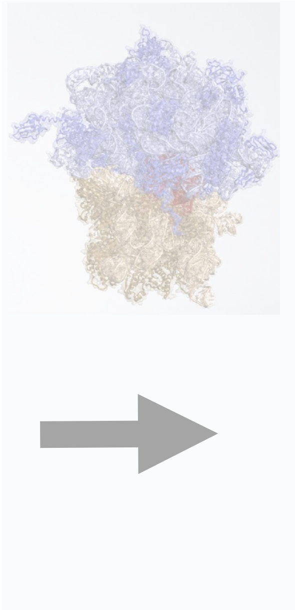
DNA





RNA



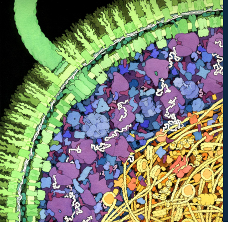


Protein

The storehouse of genetic information

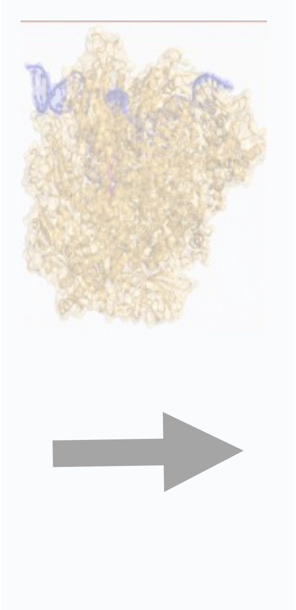
Genetic messengers

The little machine



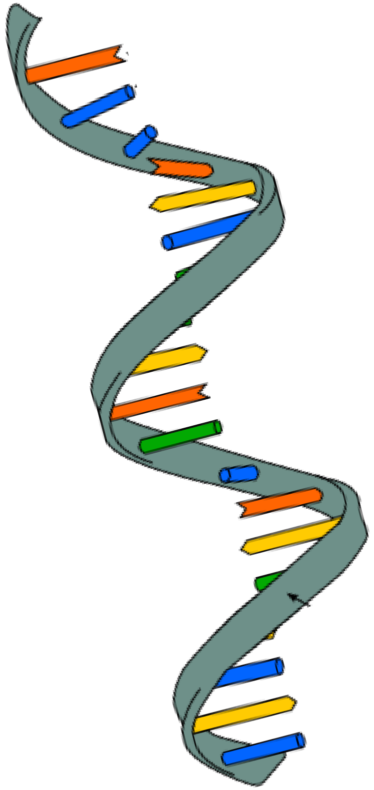
The “Central Dogma” of molecular biology

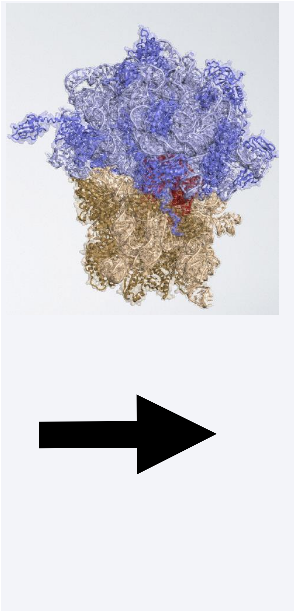
DNA



The storehouse of genetic information

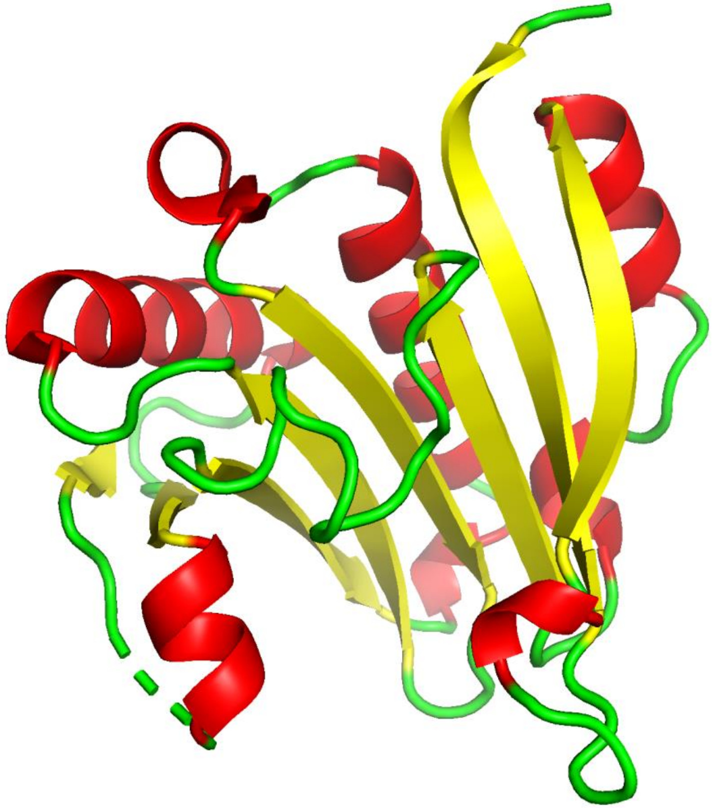
RNA



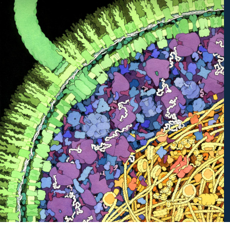


Genetic messengers

Protein



The little machine

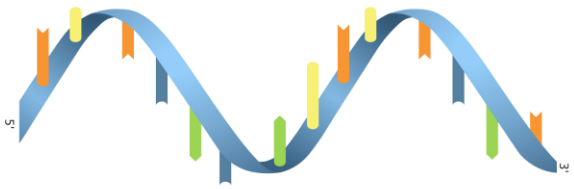


Translation

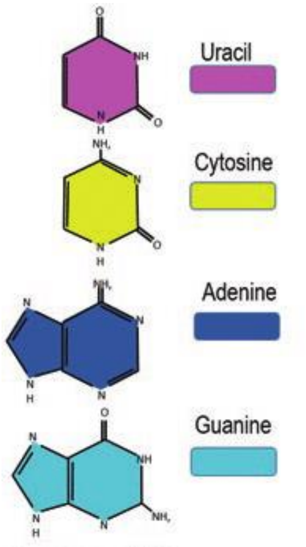
Frontiers of Science

INPUT POLYMER

Single-stranded mRNA



MADE UP OF



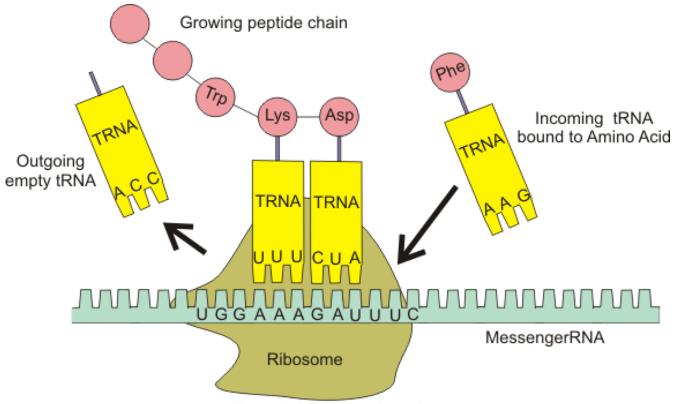
RNA nucleotides
A, C, G, U

MOLECULAR MACHINE

Ribosome

MADE OF

Protein and RNA

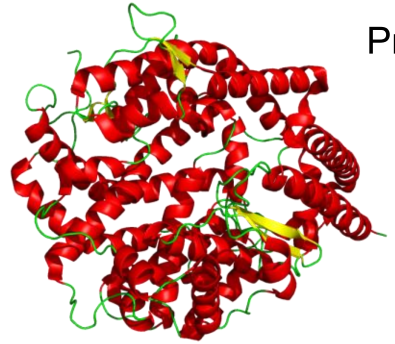


Peptide Synthesis

CONVERSION RULE

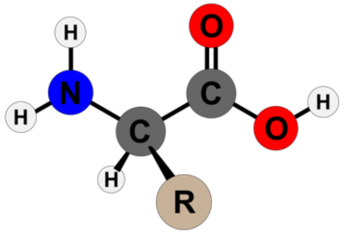
Codon-based genetic code
3 base pairs in RNA = 1 amino acid in protein

OUTPUT POLYMER

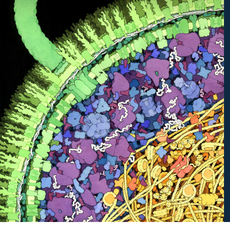


Protein

MADE UP OF

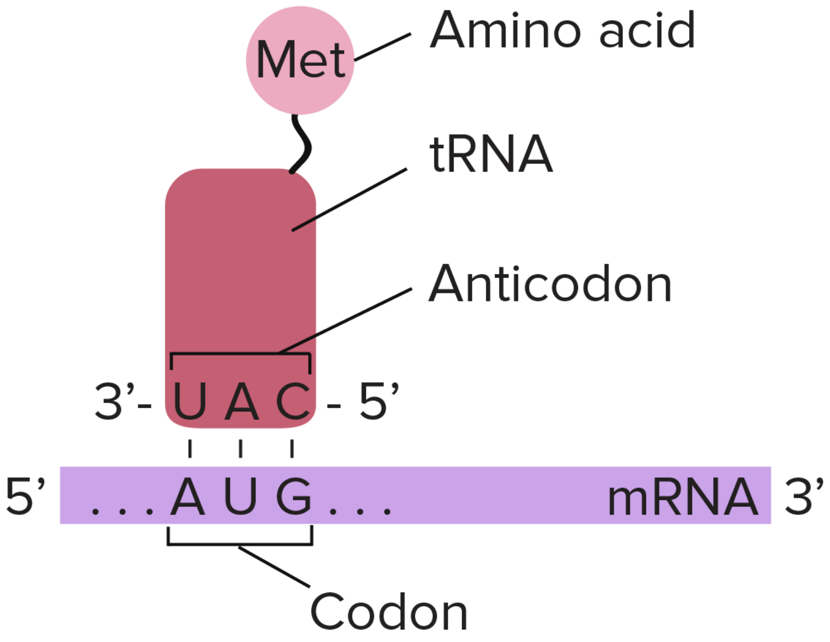


Amino acids (20)



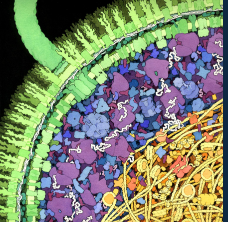
The genetic code

The genetic information in DNA and RNA is written in 3-nucleotide units called **codons** that each correspond to one amino acid



Second Position

First Position	U	C	A	G	Third Position
U	UUU, UUC: F Phe UUA, UUG: L Leu	UCU, UCC, UCA, UCG: S Ser	UAU, UAC: Y Tyr UAA: STOP UAG: STOP	UGU, UGC: C Cys UGA: STOP UGG: Trp W	U C A G
C	CUU, CUC, CUA, CUG: L Leu	CCU, CCC, CCA, CCG: P Pro	CAU, CAC: H His CAA, CAG: Q Gln	CGU, CGC, CGA, CGG: R Arg	U C A G
A	AUU, AUC, AUA: I Ile AUG: M Met	ACU, ACC, ACA, ACG: T Thr	AAU, AAC: N Asn AAA, AAG: K Lys	AGU, AGC: S Ser AGA, AGG: R Arg	U C A G
G	GUU, GUC, GUA, GUG: V Val	GCU, GCC, GCA, GCG: A Ala	GAU, GAC: D Asp GAA, GAG: E Glu	GGU, GGC, GGA, GGG: G Gly	U C A G



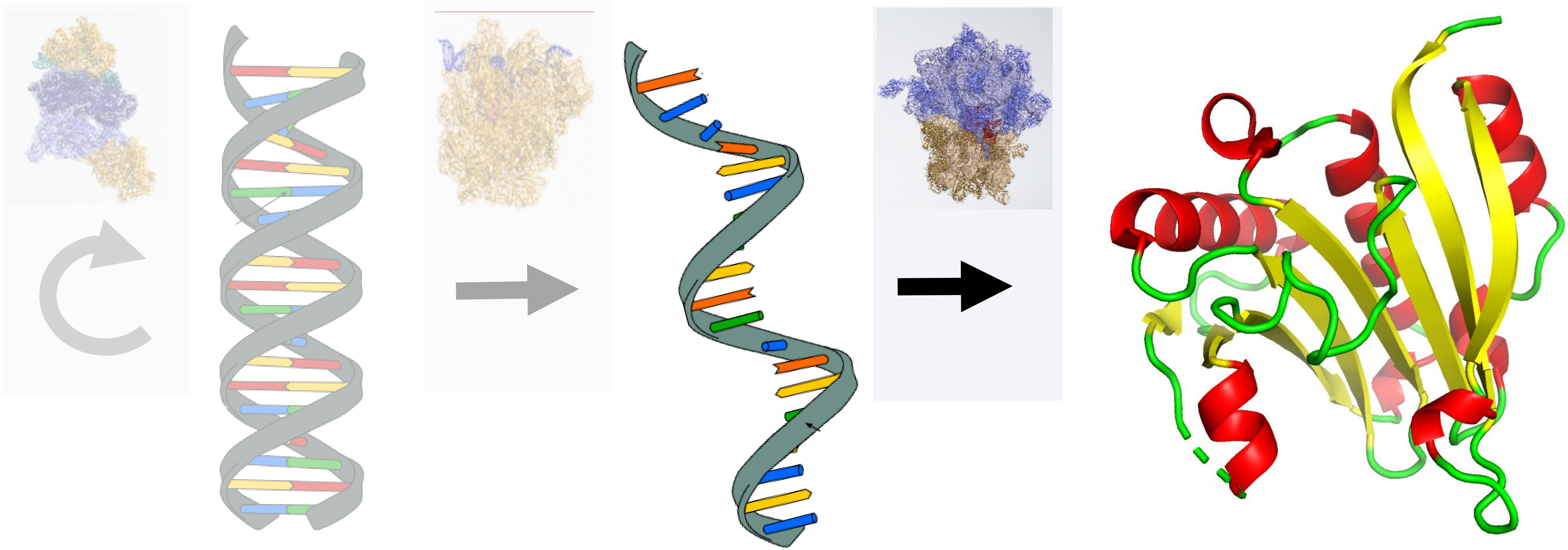
The “Central Dogma” of molecular biology

Frontiers of Science

DNA

RNA

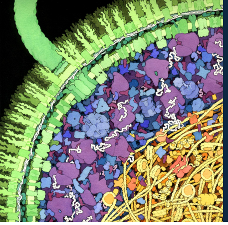
Protein



The storehouse of genetic information

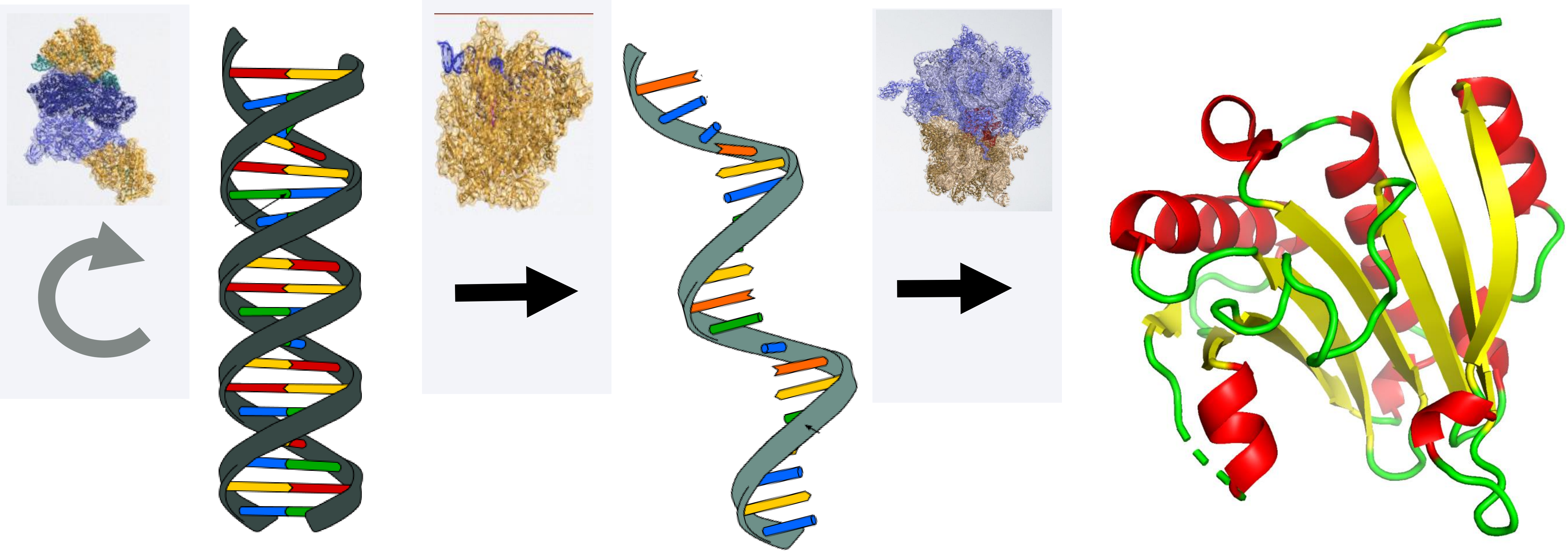
Genetic messengers

The little machine



The “Central Dogma” of molecular biology

DNA | **RNA** | **Protein**



The storehouse of genetic information

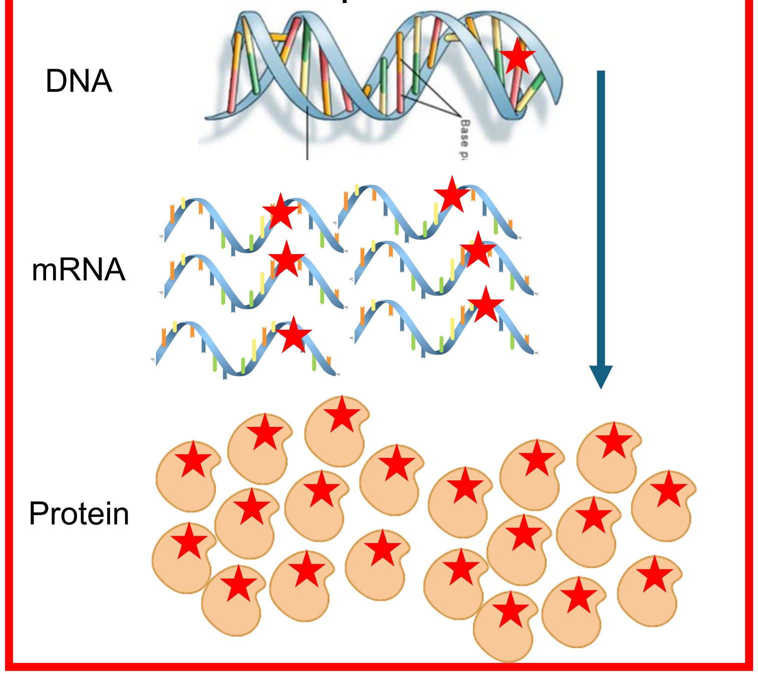
Genetic messengers

The little machine

Mistakes

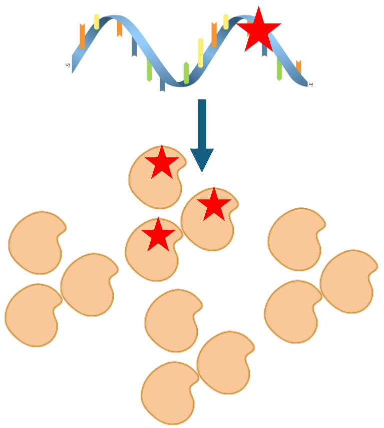
Ultimate source of all genetic variation

Replication



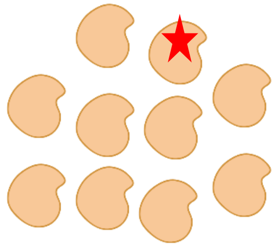
- Permanent changes to gene
- Affects all resulting proteins
- Potentially heritable to offspring

Transcription

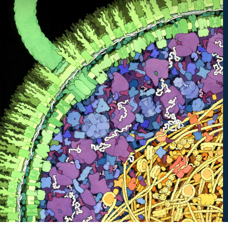


- Non-heritable changes to an mRNA transcript
- Affects a small number of proteins among many

Translation



- One-off error only affecting a single protein molecule



ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.

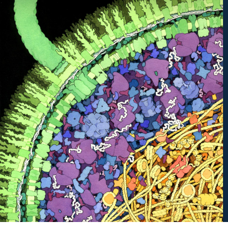
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.



How do we understand machines?

- What do we need to know to fully understand how a machine works?
  - What input it takes
  - What operation it performs
  - What output it generates
  - What components it has

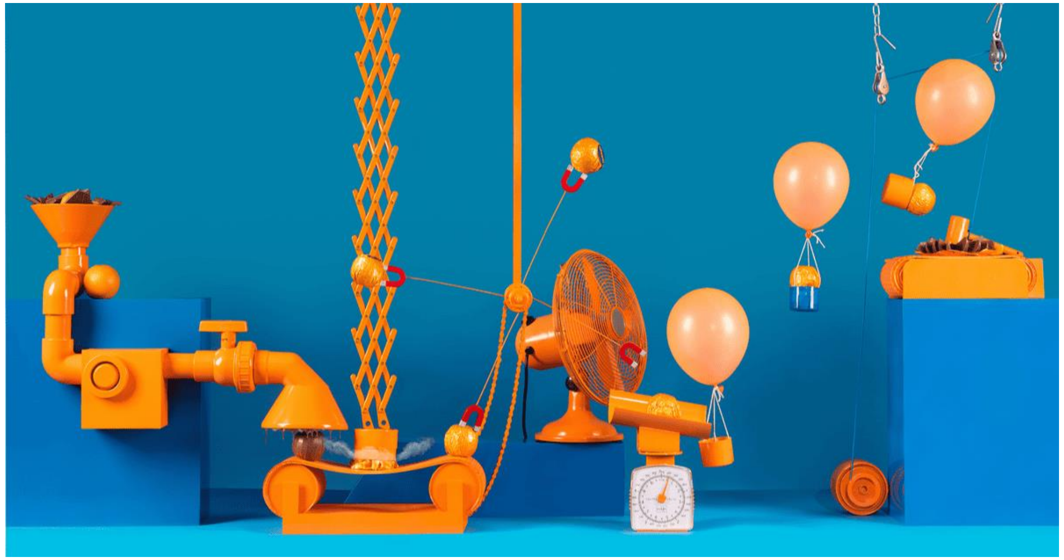
What we just covered!

  - How the components are organized
  - Where the input goes in
  - Where the output comes out

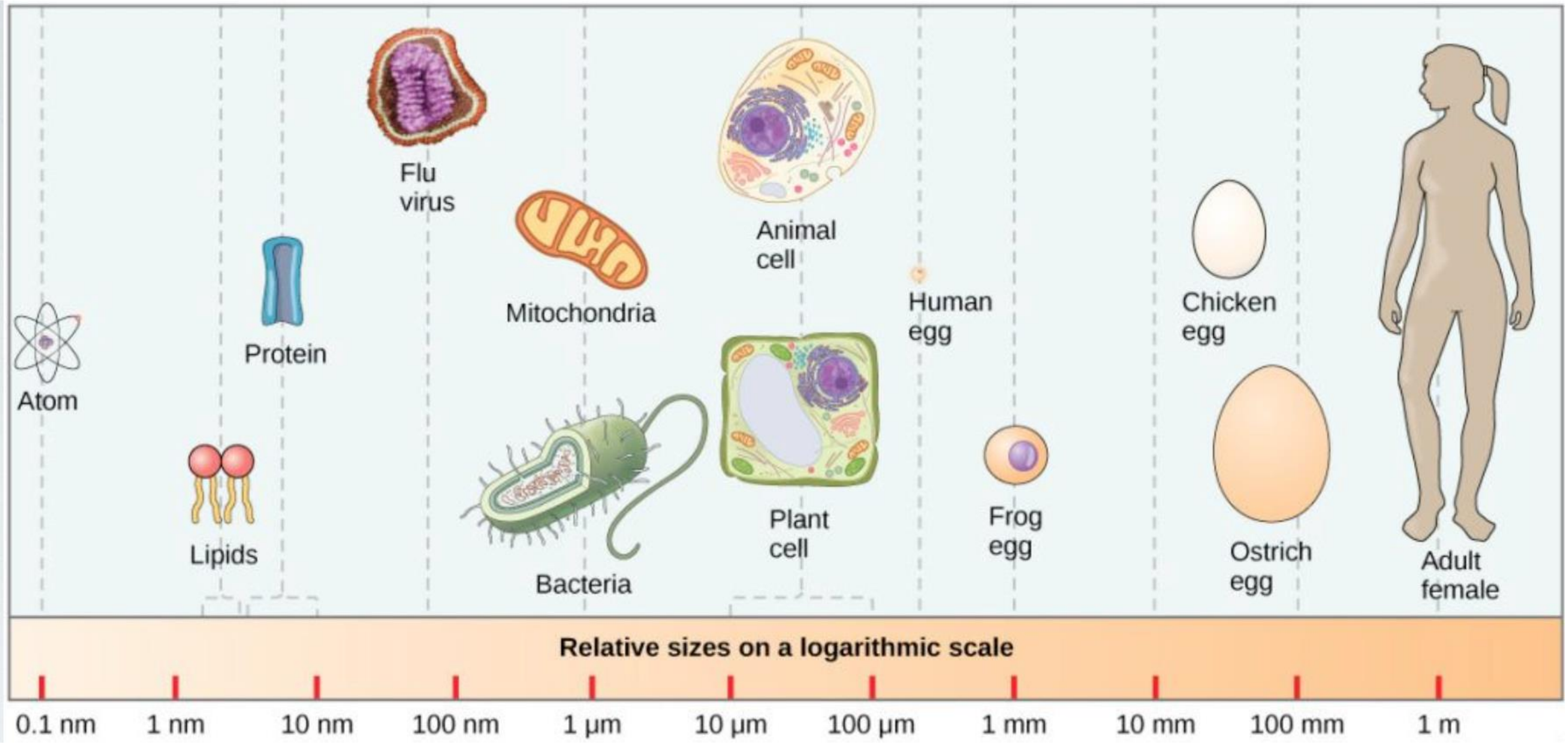
Structure

  - How it moves
  - How the components interact
  - How the input interacts with it
  - How fast it moves
  - How and where mistakes happen

Dynamics

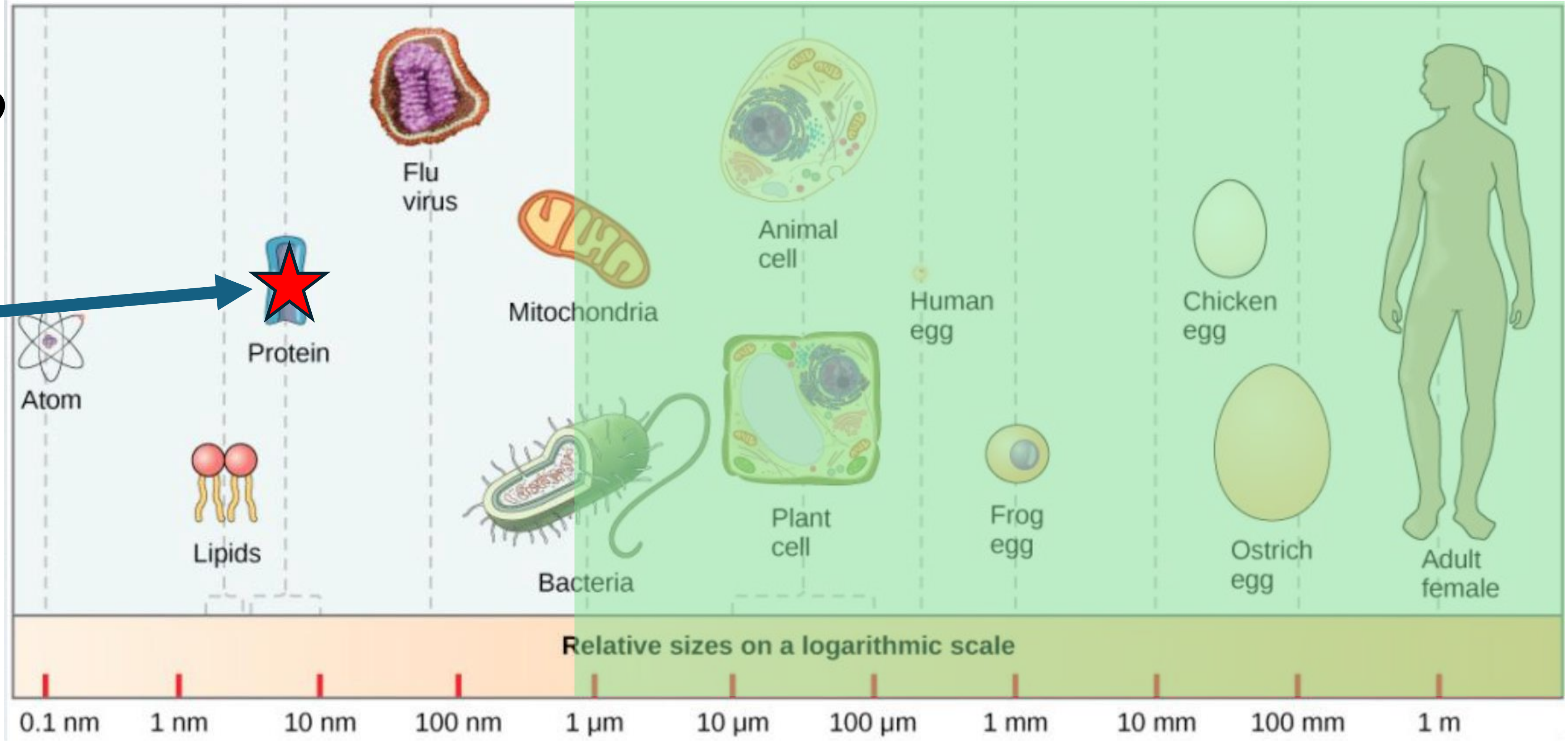


Spatial scales in chemistry and biology



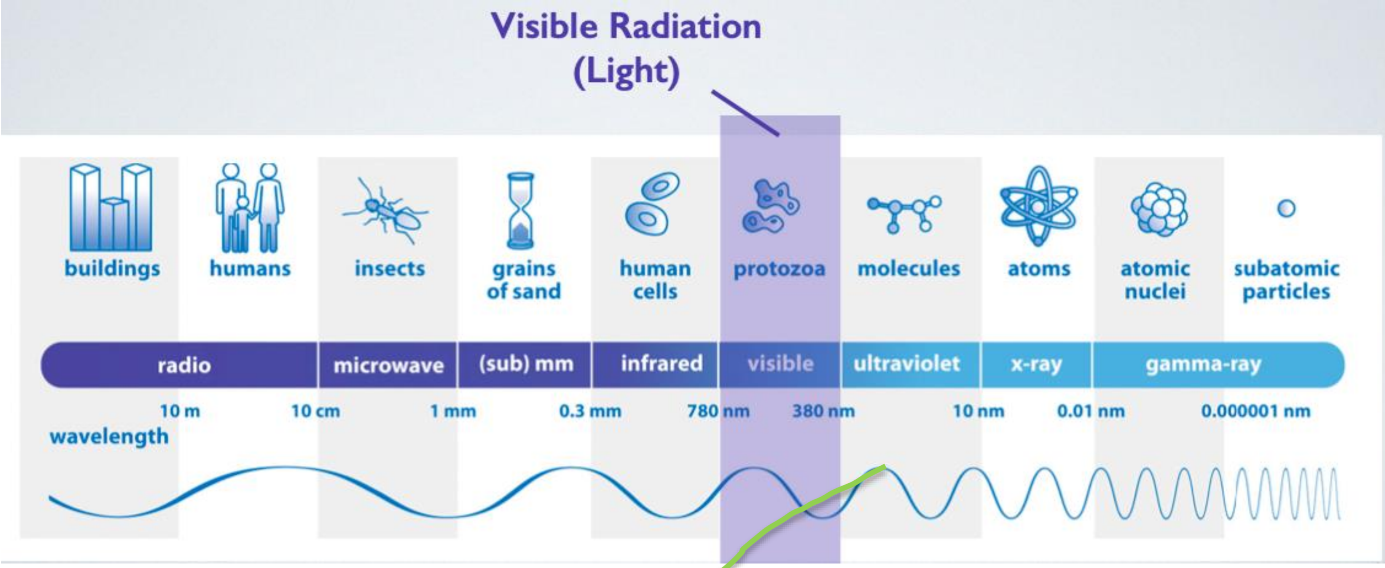
Visible light can only take you so far

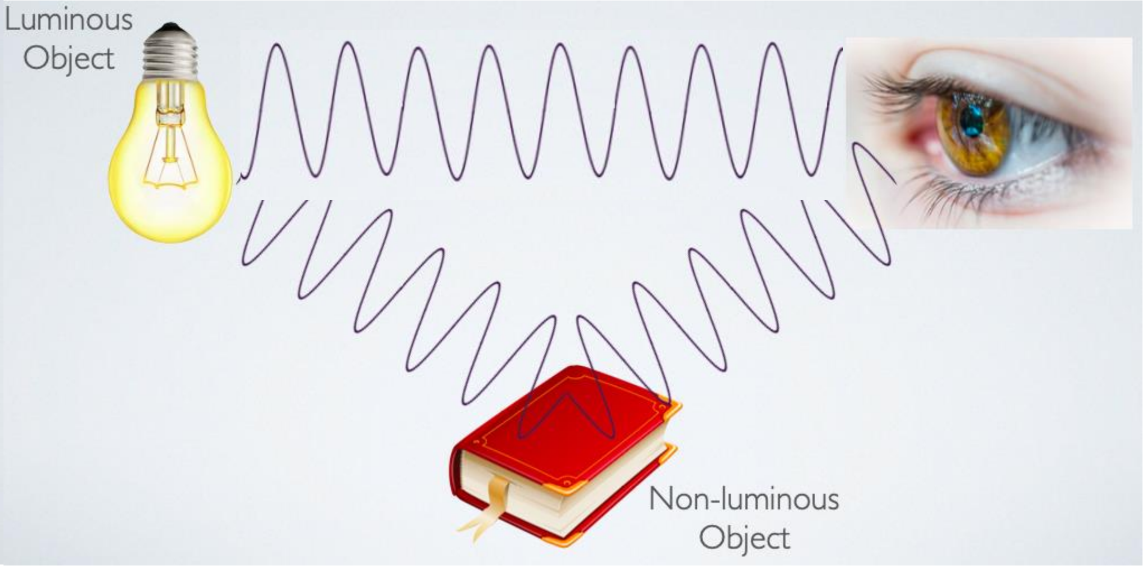
Proteins are here



How does light allow us to see objects?

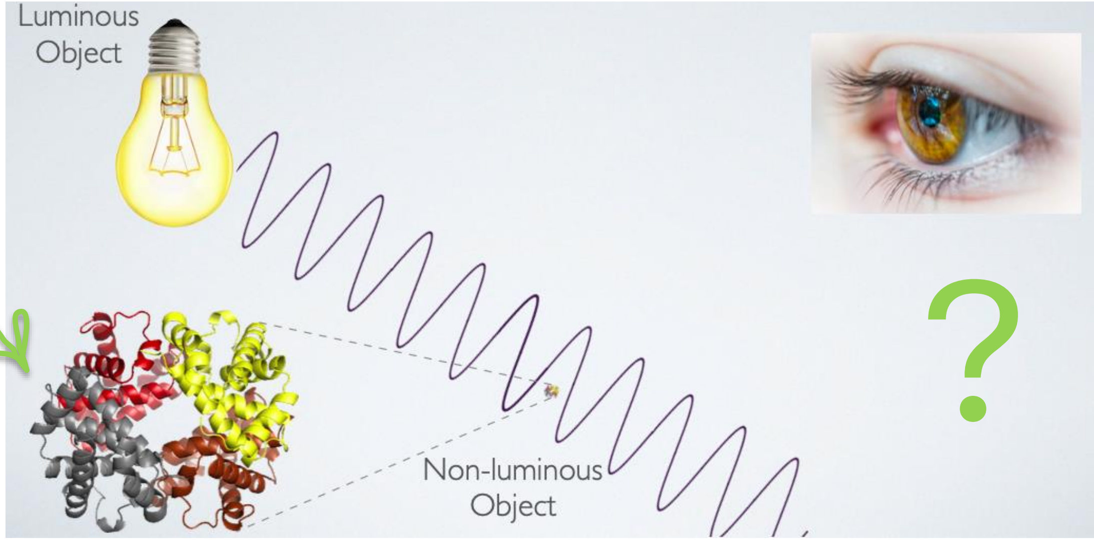
Frontiers of Science





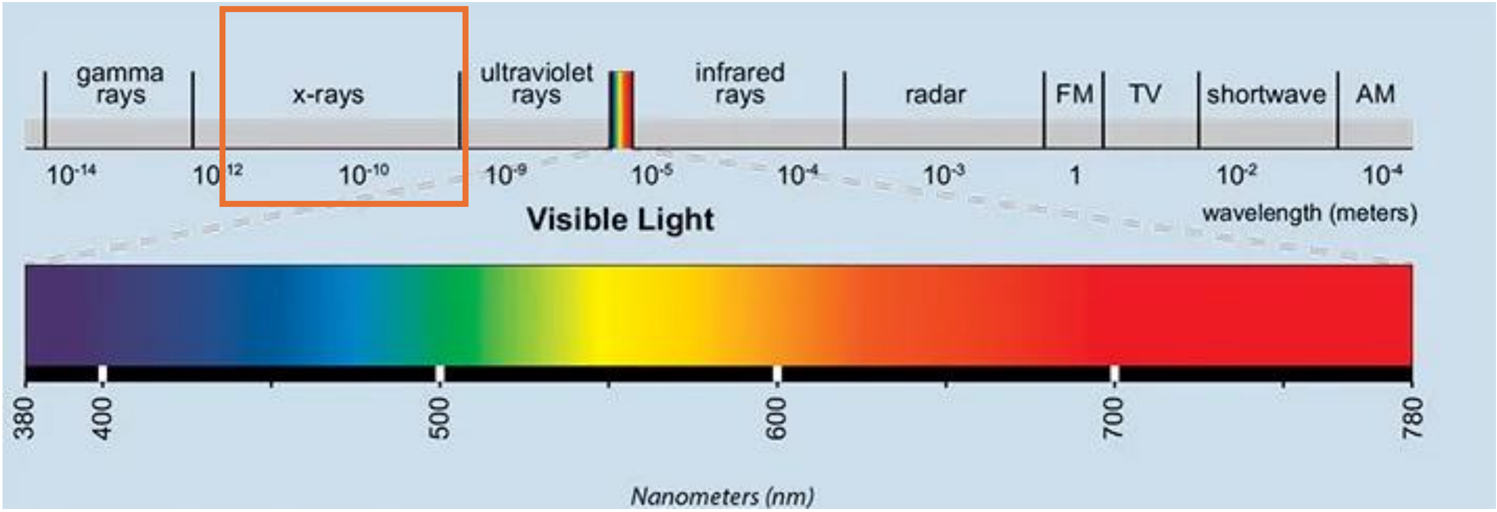
Electromagnetic radiation

Waves again!

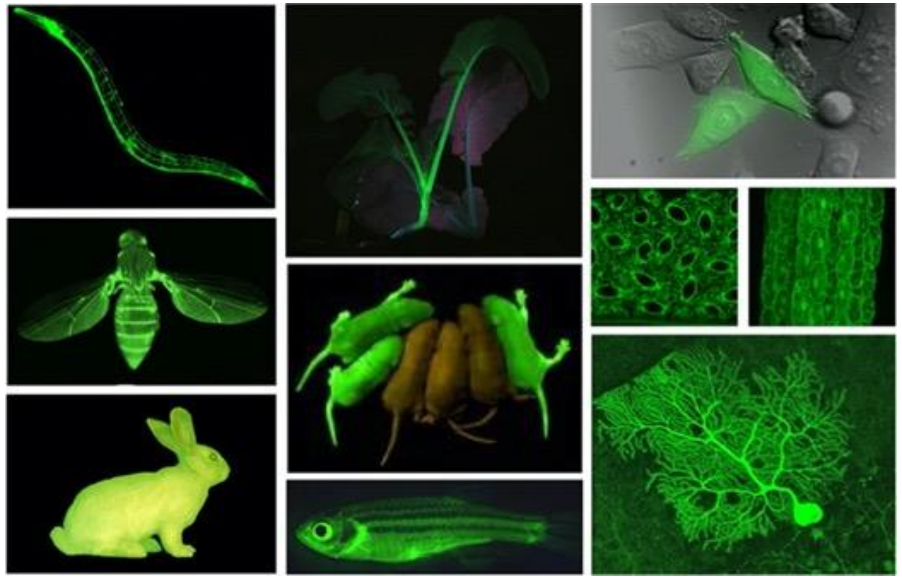


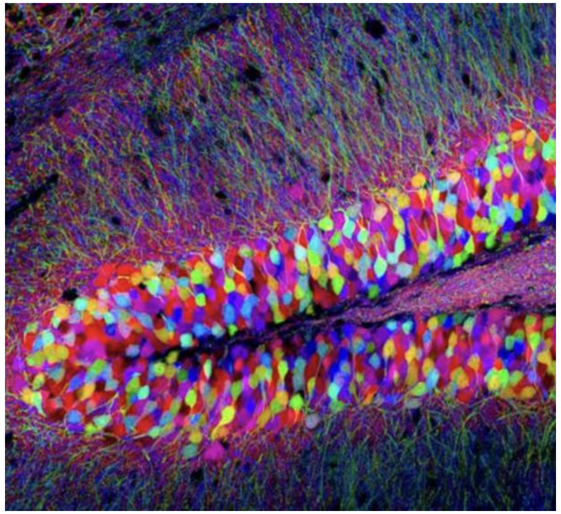
Two solutions

1) Use radiation with smaller wavelength to ”see”



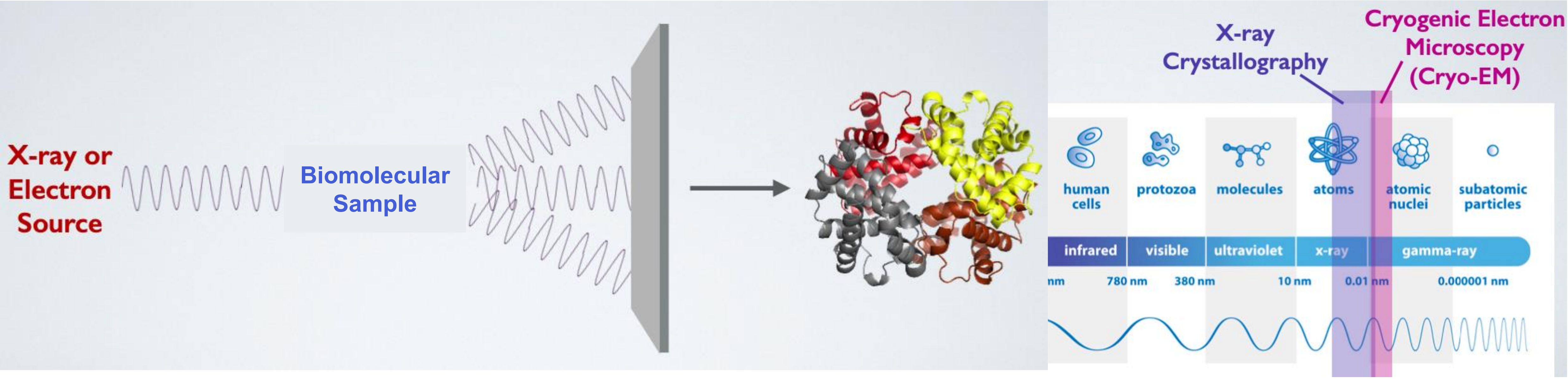
2)Put a glowing “tag” on the thing you’re looking for





X-ray crystallography and cryo-EM

Shorter-wavelength light (or electrons) resolves smaller structures

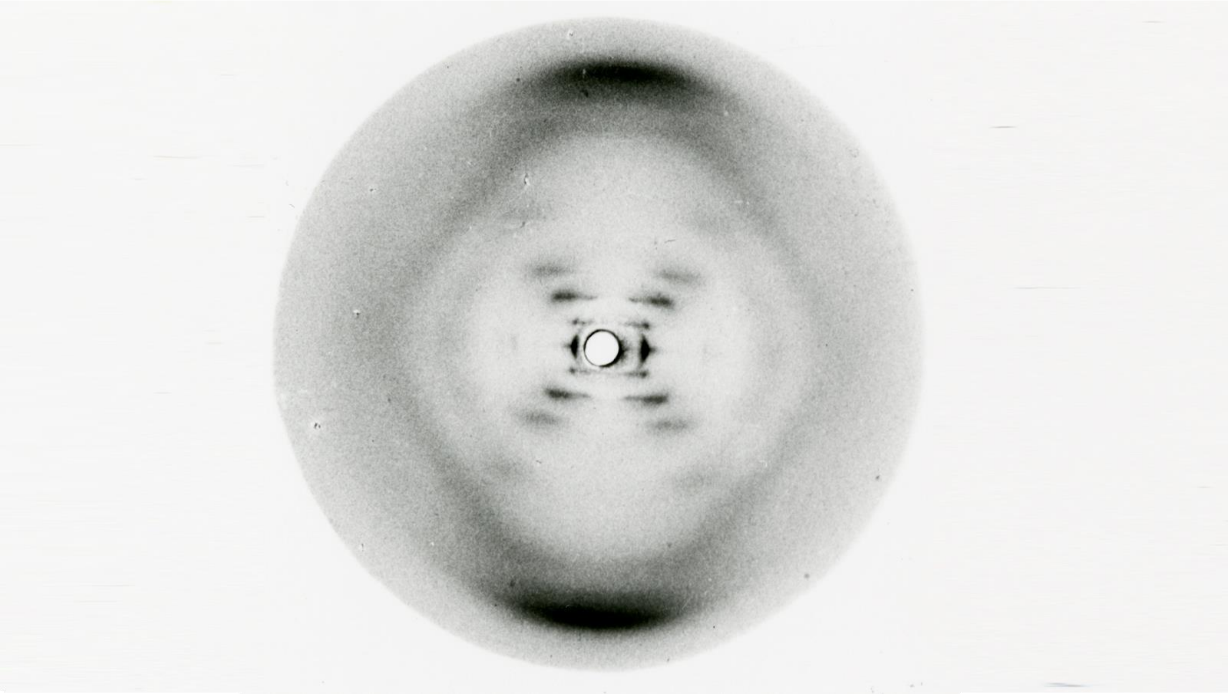


Remember - electrons behave like waves and can be used!

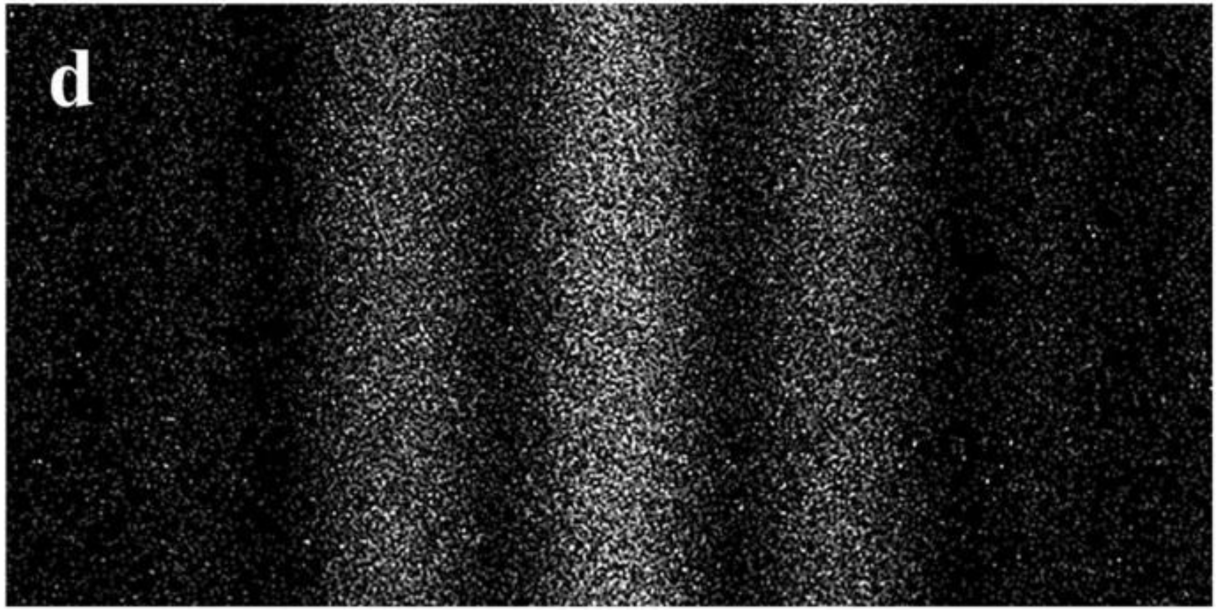
Rosalind Franklin used it in her famous Photo 51 -


from going

An example: discovering the structure of DNA



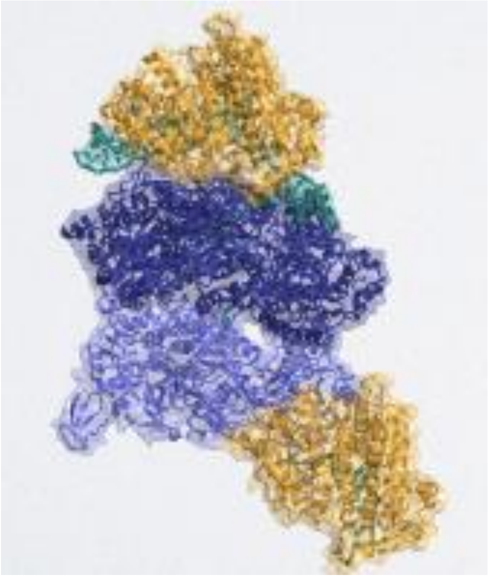
Rosalind Franklin’s “Photo 51”



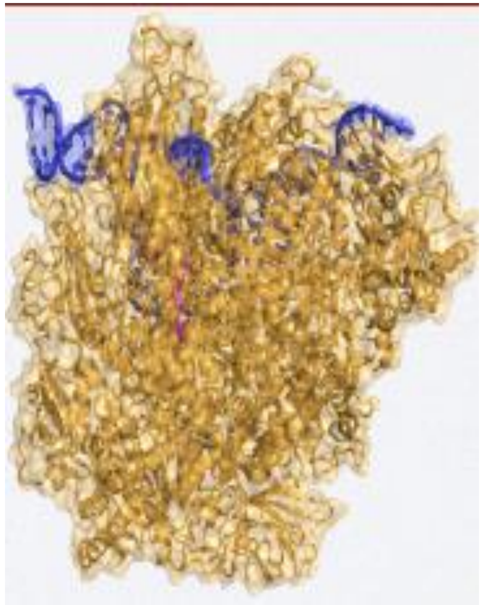
An electron double-slit experiment

Structures are super useful!

DNA polymerase



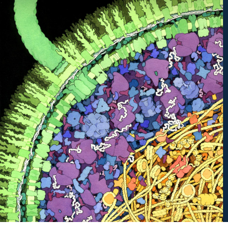
RNA polymerase



Ribosome

For these molecular machines, structures tell us:

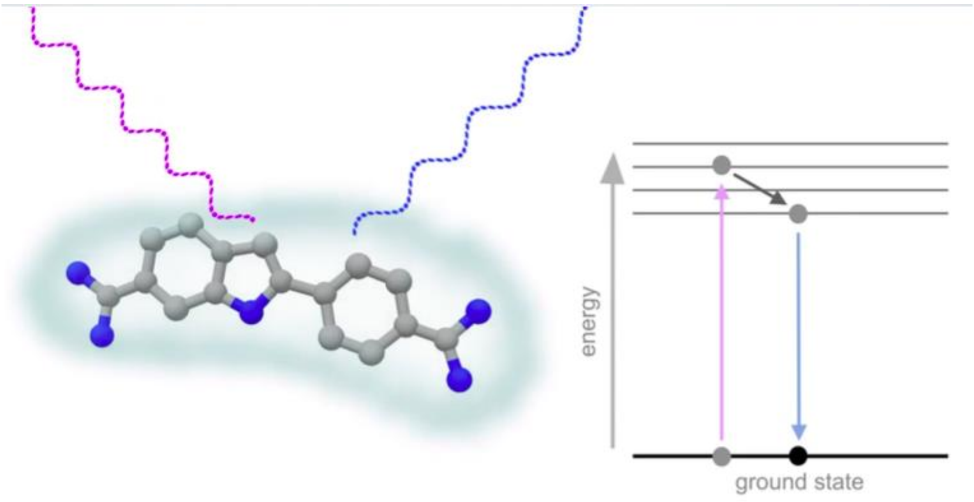
- How the components are organized
- Where the input goes in
- Where the output comes out



Biomolecular dynamics using visible radiation

Fluorescence: the emission of light by a substance that has absorbed light.

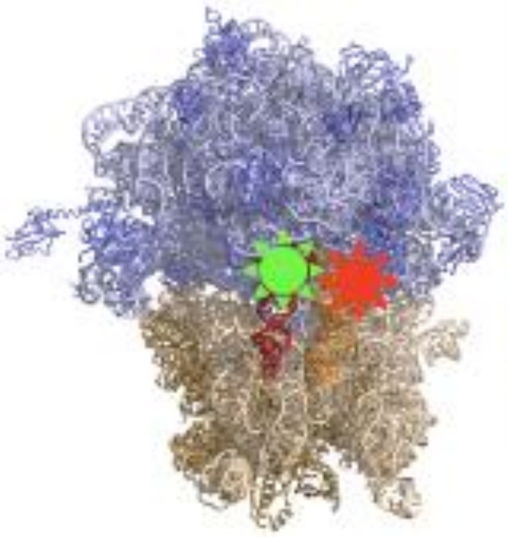
The emitted light has a different wavelength than the "incident" light.

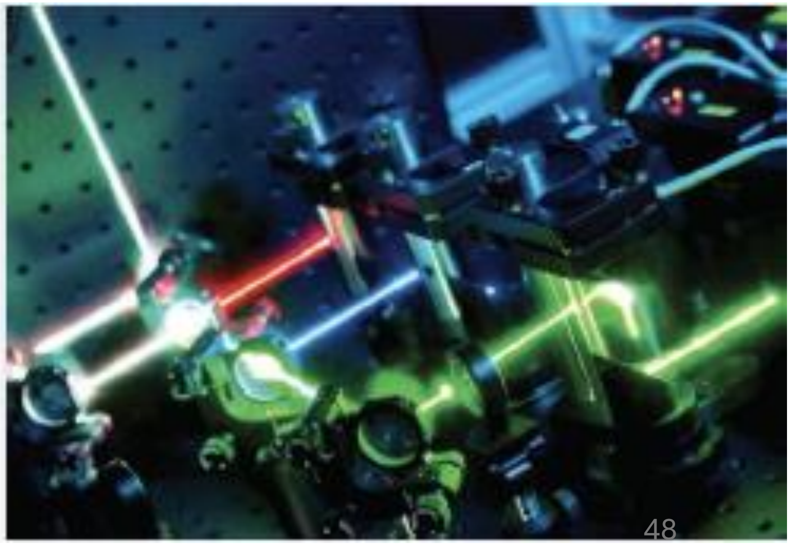


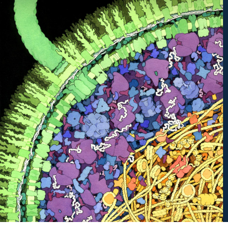
We use this to track the location, folding, and interaction of macromolecules.

For molecular machines, dynamics tell us:

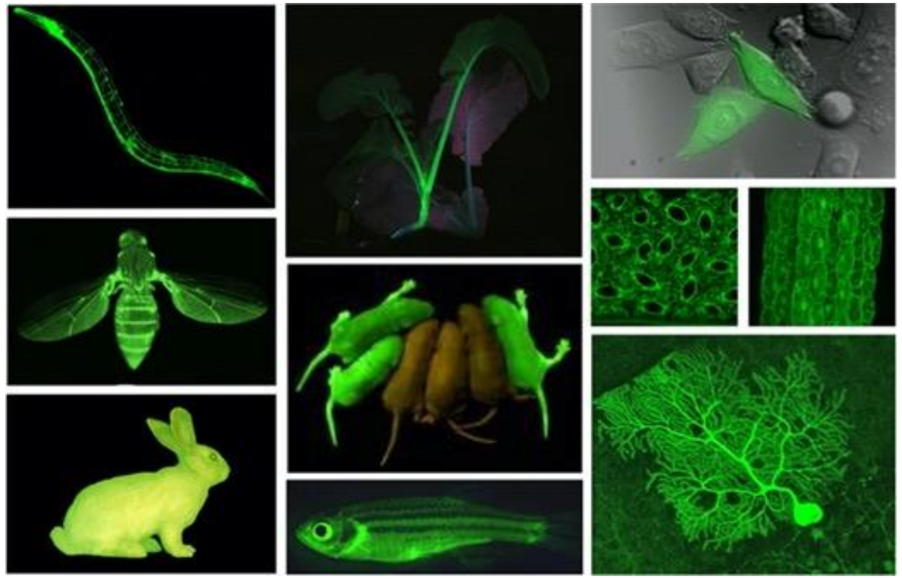
- How they move
- How the components interact
- How the input interacts with them
- How fast they move



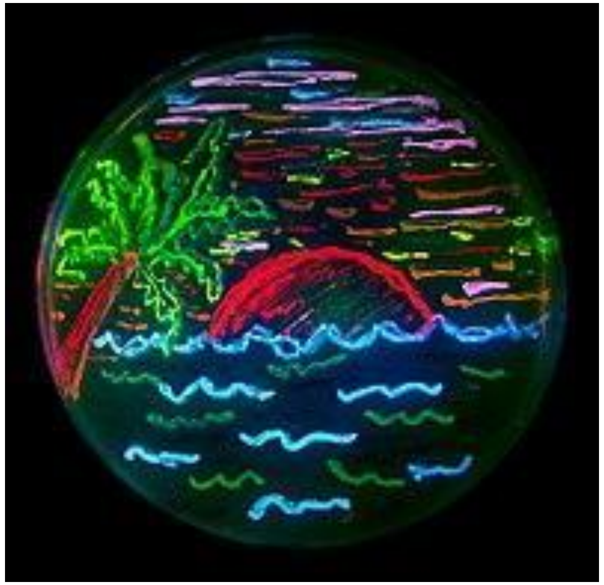




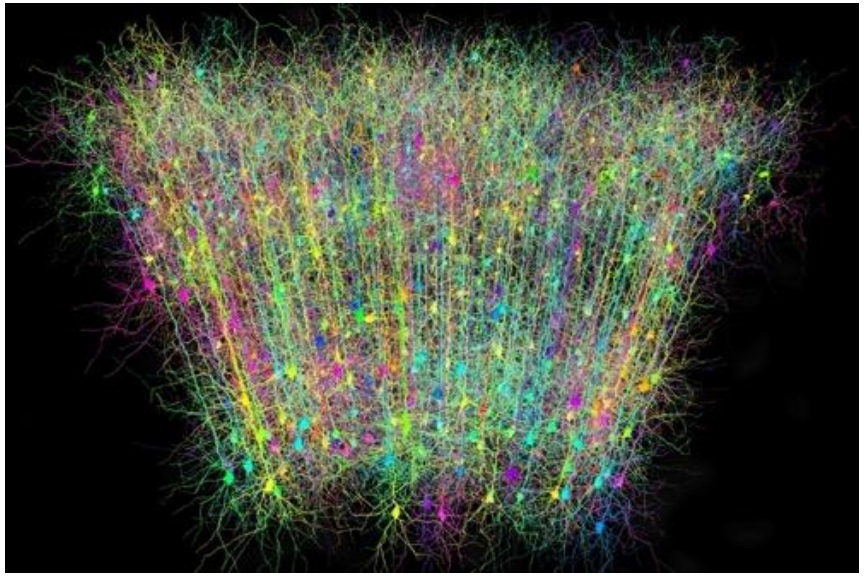
Using fluorescence to study macromolecules



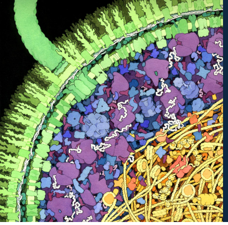
GFP - Green Fluorescent Protein taken from jelly fish emits a green light.
Prof Chalfie from CU got the Nobel Prize for GFP applying in roundworm.



Other colors of fluorescent proteins followed. Labeling different bacteria with different colors of fluorescent proteins, you can make art!



Brainbow: brain cells expressing combinations of florescent proteins with different colors. This way we can trace cells.



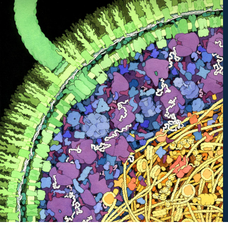
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.



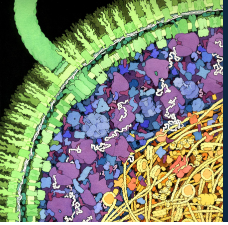
ML1 Learning Objectives

1. Develop a working vocabulary of key terms in: (1) Biophysical chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); and (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

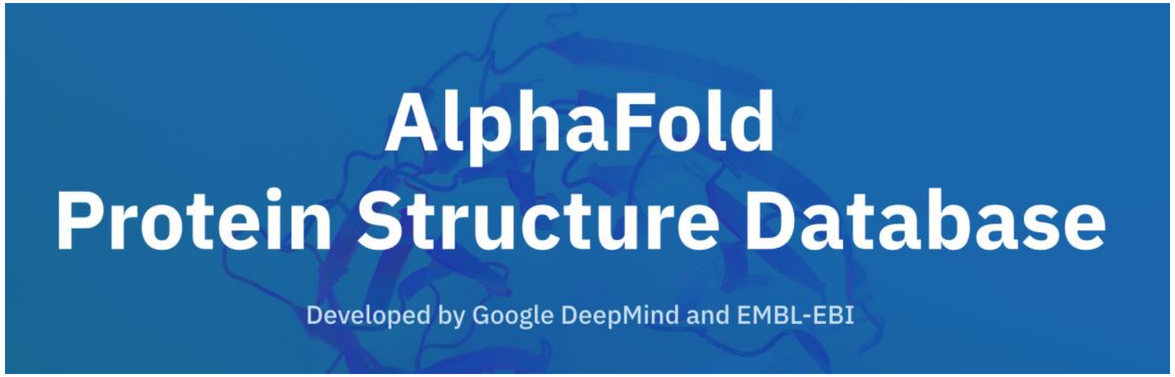
3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL and Alphafold, to access publicly available databases in efforts to understand diseases and how to combat them.



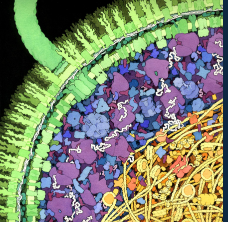
Putting it all together: AlphaFold activity!

- Goal: explore the production, folding, and misfolding of a protein involved in Alzheimer’s Disease using AlphaFold
- Form groups of ~3
- Answer questions A-D for Part 1 as a group!



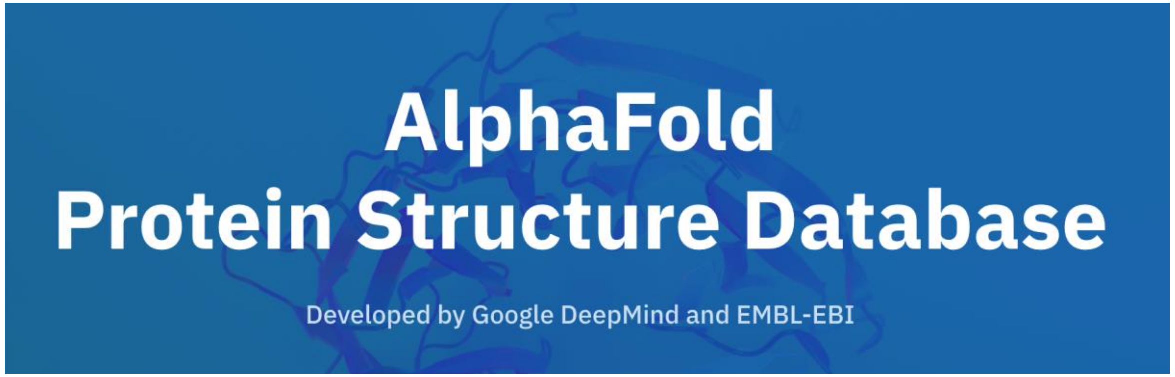
AB40 sequence:

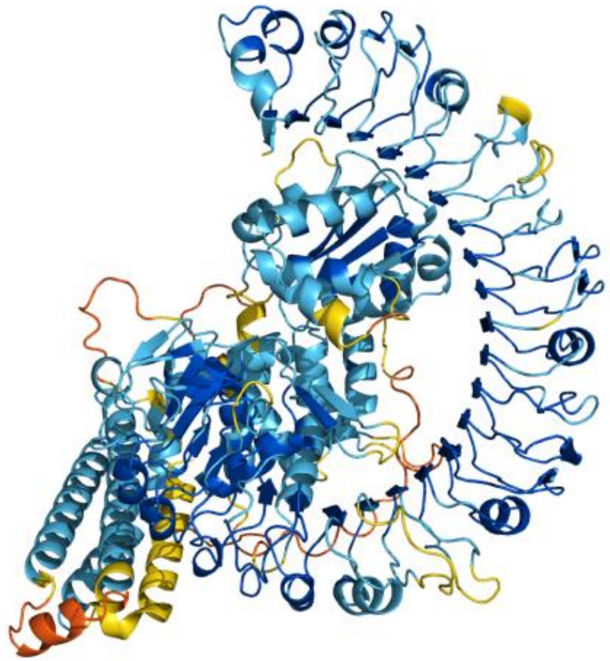
DAEFRHDSGYEVHHQKLVFFAEDVGSNKGAIIGLMVGGVV



Putting it all together: AlphaFold activity!

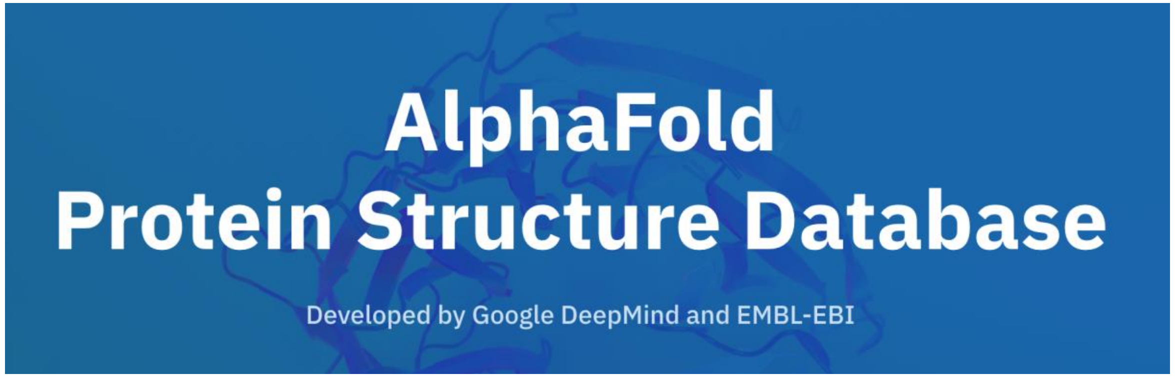
- Part 2: Protein folding
- We'll mimic the process of folding proteins after they are translated using AlphaFold
- What kind of data is used to train the models in AlphaFold?
  - AlphaFold is trained on protein structures that are solved experimentally using techniques like x-ray crystallography and cryo-EM
  - The models compare the new sequence to the sequence of proteins in the database to predict what the structure for that new protein will look like
- Let AlphaFold predict the structure of your translated protein!

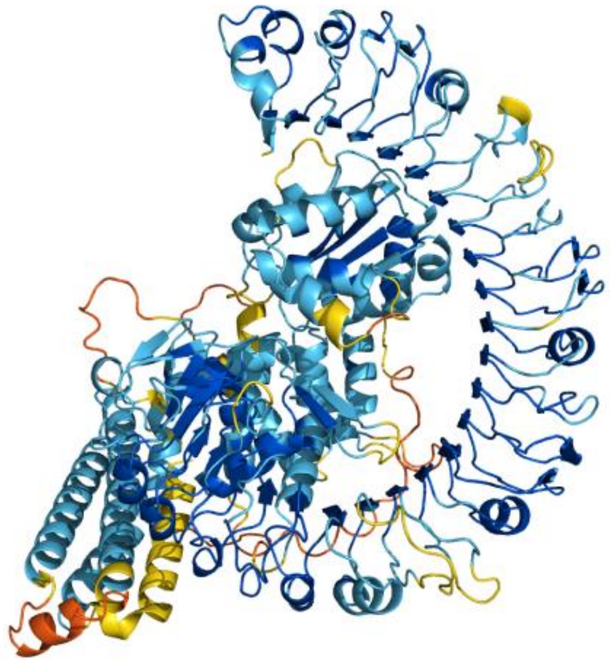


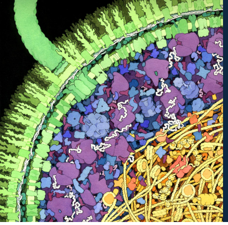


Putting it all together: AlphaFold activity!

- Part 3: Mutations, mutations, mutations
- Some cases of Alzheimer's originate from mutations in the AB40 gene
- Use your knowledge of the Central Dogma and AlphaFold to explore the effects of one of these mutations!

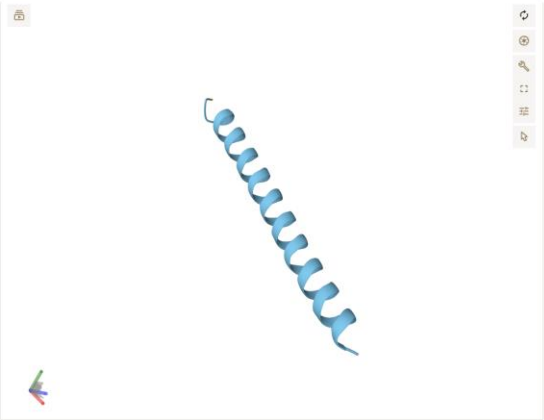






Protein folding summary: 1 vs 10 copies of AB40

One copy:



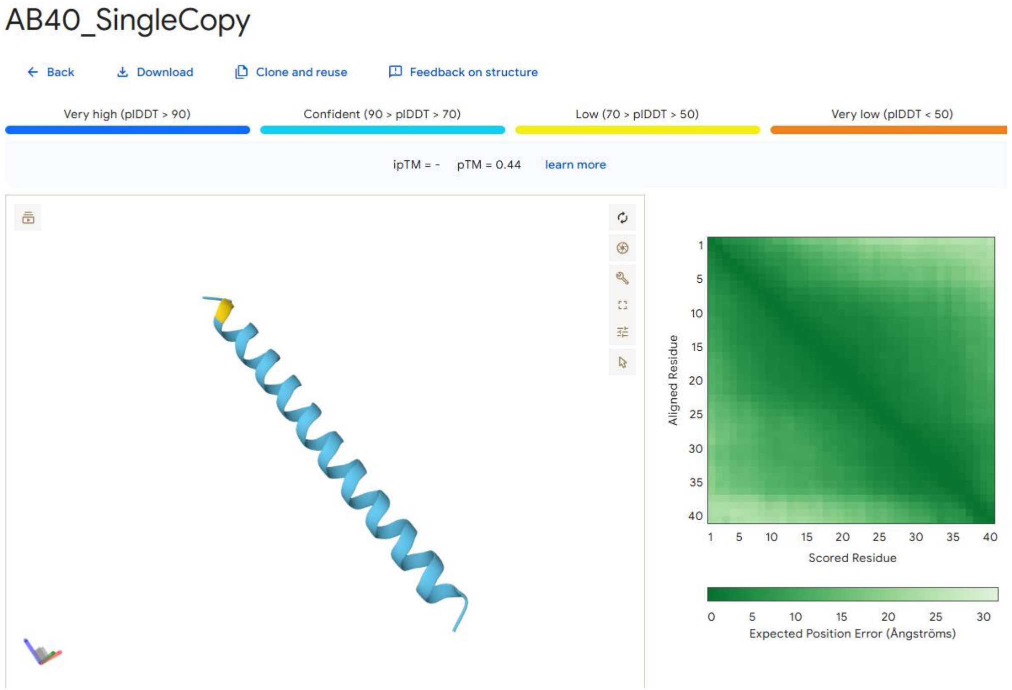
Ten copies:

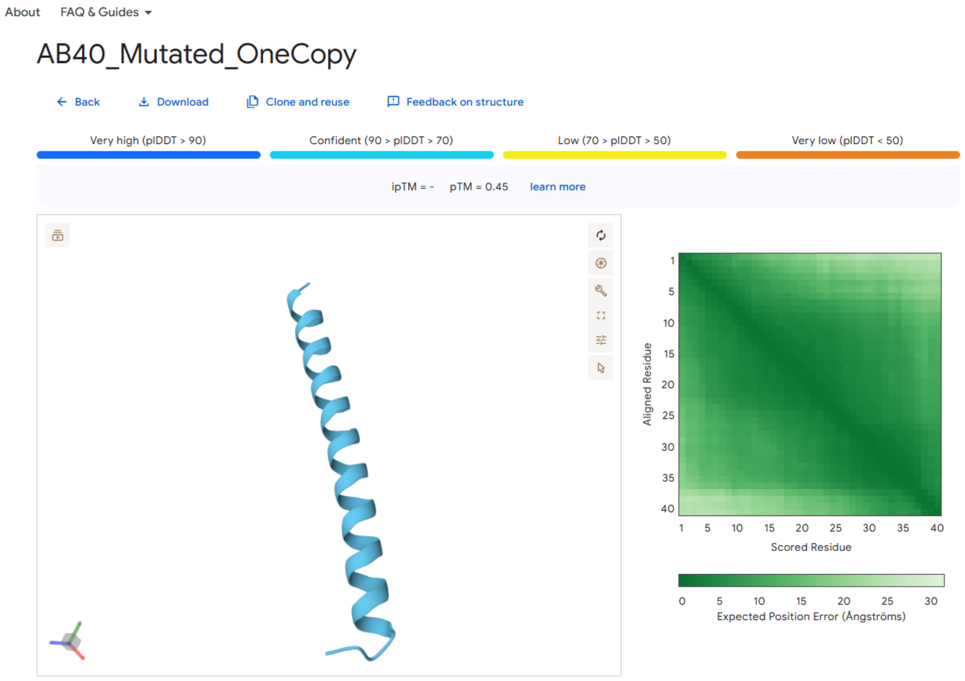
- Alpha helix held together by hydrogen bonds (within the same protein)
- Moderate confidence (not the highest, not the lowest!)

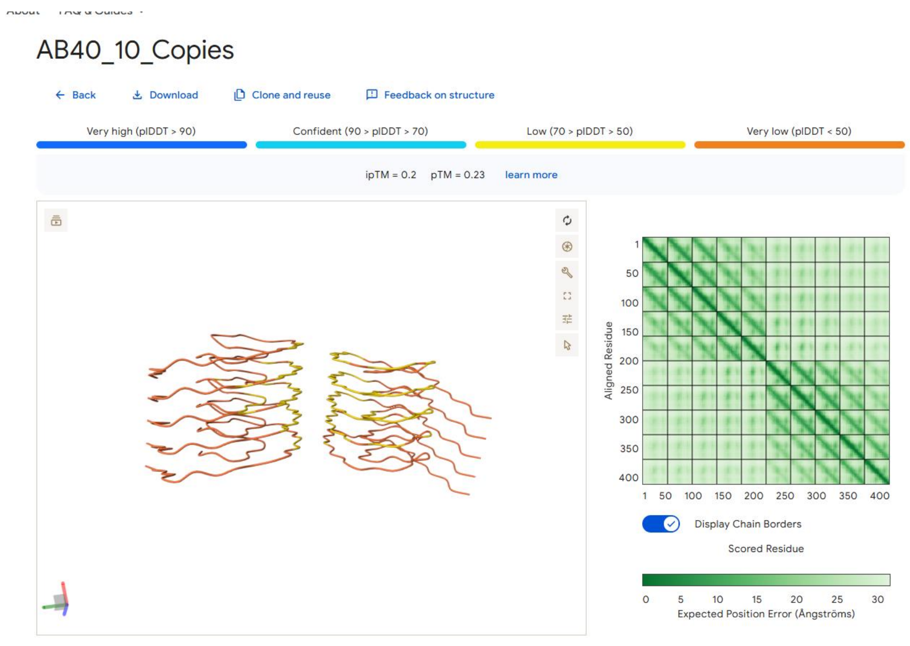
- Fibril held together by hydrogen bonds (between copies of the protein)
- Lower confidence: suggests AlphaFold may not be able to accurately predict fibril structure

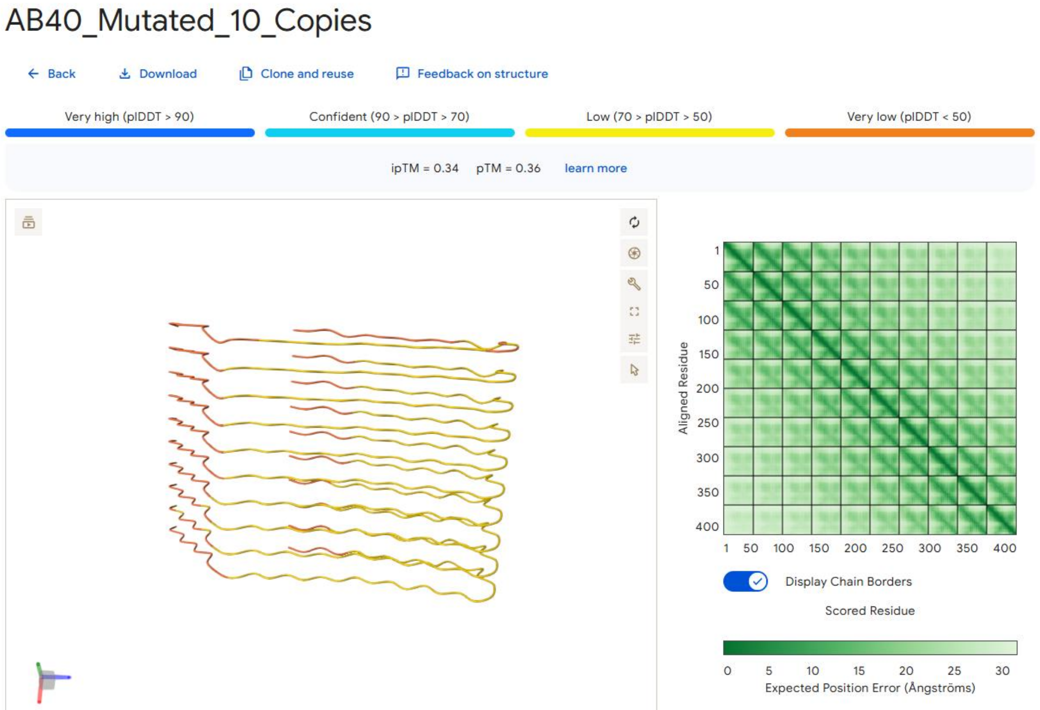
Why are these structure predictions helpful?

Even if they are not perfectly accurate, these structures help identify differences between the starting structure (in healthy neurons) and the fibril structure (in diseased neurons) that can generate predictions about how fibrils form


AB40_SingleCopy
Back
Download
Clone and reuse
Feedback on structure
Very high (pIDDT > 90)
Confident (90 > pIDDT > 70)
Low (70 > pIDDT > 50)
Very low (pIDDT < 50)
ipTM = -
pTM = 0.44
learn more
Aligned Residue
Scored Residue
Expected Position Error (Ångströms)

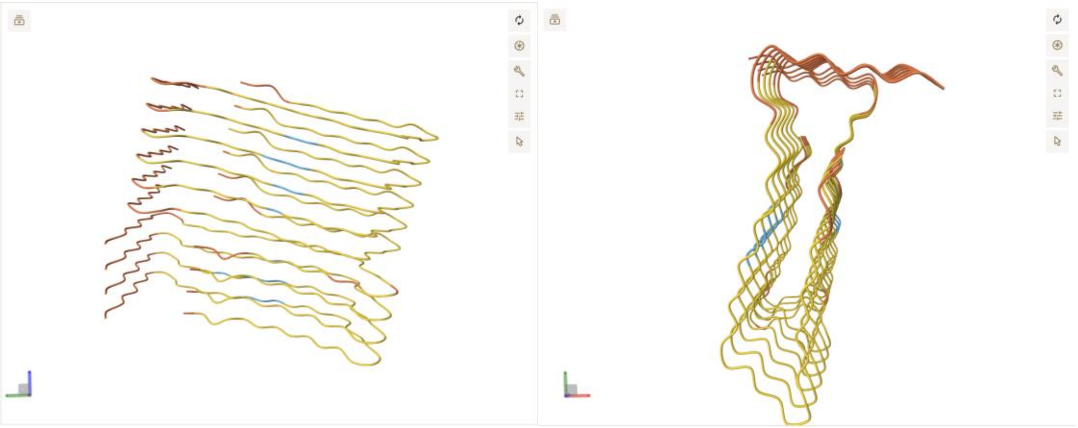

About
FAQ & Guides
AB40_Mutated_OneCopy
Back
Download
Clone and reuse
Feedback on structure
Very high (pIDDT > 90)
Confident (90 > pIDDT > 70)
Low (70 > pIDDT > 50)
Very low (pIDDT < 50)
ipTM = -
pTM = 0.45
learn more
Aligned Residue
Scored Residue
Expected Position Error (Ångströms)


AB40_10_Copies
Back
Download
Clone and reuse
Feedback on structure
Very high (pIDDT > 90)
Confident (90 > pIDDT > 70)
Low (70 > pIDDT > 50)
Very low (pIDDT < 50)
ipTM = 0.2
pTM = 0.23
learn more
Aligned Residue
Display Chain Borders
Scored Residue
Expected Position Error (Ångströms)


AB40_Mutated_10_Copies
Back
Download
Clone and reuse
Feedback on structure
Very high (pIDDT > 90)
Confident (90 > pIDDT > 70)
Low (70 > pIDDT > 50)
Very low (pIDDT < 50)
ipTM = 0.34
pTM = 0.36
learn more
Aligned Residue
Display Chain Borders
Scored Residue
Expected Position Error (Ångströms)

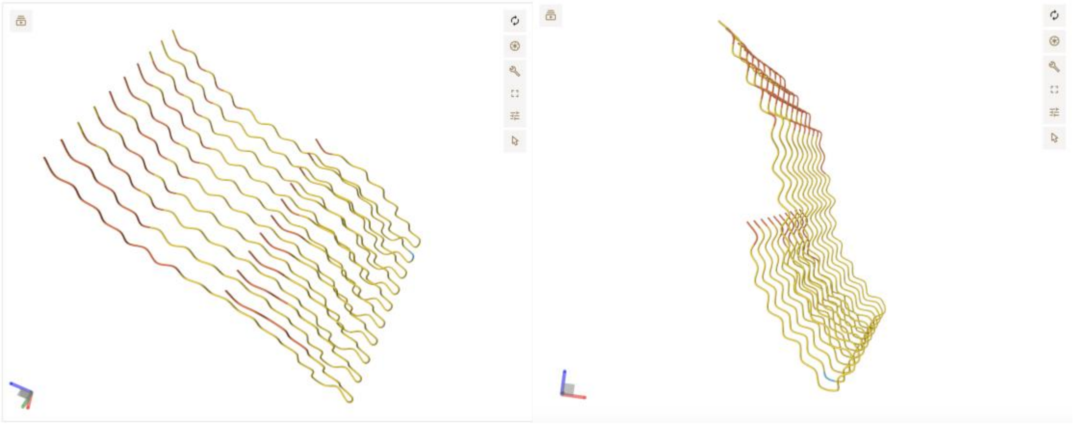
Mutations summary: unmutated vs mutated AB40

Ten copies of unmutated AB40:



- Fibril held together by hydrogen bonds (between copies of the protein)
- Lower confidence: suggests AlphaFold may not be able to accurately predict fibril structure

Ten copies of mutated AB40:



- Fibril held together by hydrogen bonds (between copies of the protein)
- Different overall fibril structure!
- Similar low confidence as unmutated fibril

What does this tell you about how individual amino acids affect the structure of proteins?

They're SUPER important! Individual amino acids are important for making hydrogen bonds within secondary structures (like alpha helices) and for determining the overall shape of the protein (like these fibrils)

Reflections

- Why would having the structure of a fibril associated with Alzheimer's disease be helpful to scientists?

- What kinds of experiments would you want to do to follow up on these predictions?

PROBABILITY PRACTICE

- The average length of a protein is 300 amino acids.
- How many possible protein sequences of average length are there?

20^{300}

The base is the number of different options that you can use for a building block, and the exponent is the size of each combination.

How do we assign probabilities?

$$P(A) = \frac{Number\ of\ outcomes\ in\ which\ A\ happens}{Total\ number\ of\ outcomes}$$

Outcomes in which event A happens

All possible outcomes

Examples:

- Probability of getting a 4 in a single roll of a die.

  P(A) = 1/6

- Probability of drawing a Queen from a standard deck of cards.

  P(A) = 4 queens/ 52 cards = 1/13

Probability Review

Formula	Examples	
Probability of A happening once in a single try: P(A)	Probability of getting a 4 in a single roll of a die: 1/6	Probability of getting hit by a 1 Mton asteroid in a given year: P = 1/recurrence interval $1/100 = 1 \times 10^{-2}$
Probability of A not happening at all in a single try: 1 – P(A)	Probability of not getting a 4 in a single roll of a die: 1 – 1/6 = 5/6	Probability of **not** getting hit by a 1 Mton asteroid in a given year: $1 - (1 \times 10^{-2})$
Probability of A not happening at all in a n tries (see Note): $[1 - P(A)]^n$	Probability of not getting a 4 in 10 rolls of a die: $[1 - 1/6]^{10} = (5/6)^{10}$	Probability of **not** getting hit by a 1 Mton asteroid in 80 years: $[1 - (1 \times 10^{-2})]^{80}$
Probability of A happening at least once in n tries: $1 - [1 - P(A)]^n$	Probability of getting at least one 4 in 10 rolls of a die: $1 - [1 - 1/6]^{10} = 1 - (5/6)^{10}$	Probability of getting hit **at least once** by a 1 Mton asteroid in a 80 years: $1 - [1 - (1 \times 10^{-2})]^{80}$ = 0.55

Note: Calculating the combined probability for two independent events A and B:

Probability of A AND B happening = $P(A) \times P(B)$

Probability of A OR B happening = $1 - [1 - P(A)] \times [1 - P(B)]$

PROBABILITY PRACTICE

- The viral genome is 30,000 nucleotides long.
- It accumulates an average of 23.8 nucleotide substitutions per year.

What is the probability of the original viral sequence to acquire a mutation in a particular site in one year?

PROBABILITY PRACTICE

- The viral genome is 30,000 nucleotides long.
- It accumulates an average of 23.8 nucleotide substitutions per year.

What is the probability of the original viral sequence to acquire a mutation in a particular site in one year?

$23.8 / 30{,}000 = 7.9 \times 10^{-4}$

Probability of getting A in a single try: P(A)

- The probability that a random undergrad was born in NYC.

  P(A) = (total # of undergrads born in NYC) / (total # of undergrads)

- If a meteor the size of Chicxulub is coming toward Earth, what is the probability it will hit land?

  P(A) = (surface area of land)/(surface area of the Earth) = 0.3

- A “Chelyabinsk size” meteor hits the Earth on average every 100 years. What is the probability that it will hit next year?

  P(A) = 1/(recurrence interval) = 1/100

Probability of NOT getting A in a single try: 1 – P(A)

- Probability of **NOT** getting 4 in a single roll of a die.

  P(not A) = 1 – P(A) = 1 – 1/6=5/6

- Probability of **NOT** drawing a Queen from a deck of cards.

  P(not A) = 1 – P(A) = 1 – 1/13 = 12/13

- A "Chelyabinsk size" meteor hits the Earth on average every 100 years. What is the probability that it will **NOT** hit next year?

  1 – P(A) = 1 – 1/(recurrence interval) = 1 – 1/100 = 99/100

- The probability that a random undergrad is **NOT** born in NYC.

  1 – P(A) = 1 – (total # of undergrads born in NYC) / (total # of undergrads)

- If a meteor the size of Chicxulub is coming toward Earth, what is the probability it will hit water (i.e. **NOT** land)?

  1 – P(A) = 1 – (surface area of land)/(surface area of the Earth) =

  1 – 0.3 = 0.7

Probability of getting A <u>AND</u> B, where A and B are independent events:

P(A)P(B)

- Probability of getting two 4's in a row.

$$P(A \text{ AND } B) = P(A)P(B) = 1/6 \times 1/6 = 1/36$$

- Probability of drawing four Queens in a row from a deck of cards (with replacement, you put the card that you drew back in the deck).

$$[P(A)]^4 = (1/13)^4 = 3.5 \times 10^{-5}$$

- Probability of drawing all four Queens from a deck of cards (without replacement, you keep each card that you draw).

$$[P(A)]^{1\text{st card}} \times [P(A)]^{2\text{nd card}} \times [P(A)]^{3\text{rd card}} \times [P(A)]^{4\text{th card}} =$$

$$= (4/52) \times (3/51) \times (2/50) \times (1/49) = 3.6 \times 10^{-6}$$

- The probability that a random undergrad was born in NYC and is a 1st year student.

$$P(A)P(B) = (\text{total \# of undergrads born in NYC}) / (\text{total \# of undergrads}) \times 1/4$$

Probability of getting A OR B, where A and B are independent events:

- For two independent events A and B there are two possible mutually exclusive outcomes:
  - Either A or B happens (only A happens, or only B happens, or both A and B happen).
  - Neither A, nor B happens.

P = 1

- Probability of neither A and neither B happening is:

$$P(\text{not } A)\, P(\text{not } B) = [1 - P(A)]\,[1 - P(B)]$$

- Probability of A or B happening is:

$$1 - P(\text{not } A)\, P(\text{not } B) = 1 - [1 - P(A)]\,[1 - P(B)]$$

- Probability of A or B or C or D happening is:

$$1 - [1 - P(A)]\,[1 - P(B)]\,[1 - P(C)]\,[1 - P(D)]$$

Probability of getting A <u>OR</u> B, where A and B are independent events:

- What is the probability that a card chosen randomly from a 52-deck of cards will be a Jack or a heart?

P(Jack)= 4/52

P(heart) = 13/52

P(Jack or heart) = 1 – [1 – P(Jack)] [1 – P(heart)] =

= 1 – [1 – (4/52)] [1 – (13/52)] =

= 1 – (48/52)(39/52) = 16/52

Note: You can also figure this out by counting cards:

There are 13 heart cards (one if which is a Jack).

Additionally there are 3 remaining Jack cards that are not hearts.

So in total there are 13+3 = 16 cards out of 52 that are either a Jack or a heart.

Probability: 16/52

Probability of event A occurring at least once in N tries:

- In N tries there are two possible mutually exclusive outcomes:

P = 1
  - Either A happens at least once in N tries.
  - A never happens in all N tries.

- Probability of A not happening in a single try:

$$P(\text{not } A) = [1 - P(A)]$$

- Probability of A not happening in all N tries:

$$P(\text{not A in } 1^{st} \text{ try})\ P(\text{not A in } 2^{nd} \text{ try}) \ldots (P \text{ not A in } N^{th} \text{ try})$$

$$= [1 - P(A)]^N$$

- Probability of A happening at least once in N tries:

$$1 - [1 - P(A)]^N$$